Memory Without a Writer: Agent Memory Built by Code

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Code, data and protocol github.com/MemorableOrg/memory-without-a-writer

Abstract

Agent memory systems often ask a language model to decide what to remember before future questions are known. This adds ingestion cost and can discard details that later matter. We investigate whether effective memory requires this generative write step. Our approach preserves the original history and retrieves evidence when a question arrives, with no generative model calls to construct the store: conversation turns, or whole agent trajectories, are kept as they were, indexed by keyword and by a local embedding model, and reordered at question time by a fused score, two dialogue rules and two local cross-encoders. An agent pays a memory's build and read costs on every session while a benchmark score is paid once, so we report accuracy together with tokens read, search time and build cost. With GPT-5 reading the top 40 stored turns, the store answers 90.6 percent of 607 LoCoMo questions, 385 of them development questions used to tune the search, and 82.0 percent of the 222 test questions completed, at 3,294 reader tokens per question and about one second of search. Retrieval holds up without model-written memories: the store finds every gold session in the top 10 for 97.4 percent of 470 LongMemEval-S questions, including development questions used for tuning, recovers 80.3 percent of LoCoMo gold turns in the top 10 against 64.0 for a flat retrieval baseline, and ranks agent trajectories on Proced-Mem at 0.957 mean average precision. The remaining errors expose a tradeoff between preserving information and finding all the evidence needed to answer. These results motivate agent memory that retains the evidence and leaves its interpretation to the moment of use. The code, the per-question results and the protocol that registered every run are public.

1 Introduction

We propose an agent memory without a generative writer. Here, a writer is a language model that selects and rewrites past interactions for storage. Our approach retains the original history and retrieves relevant evidence when a question arrives. Agent memories make information from earlier interactions available to later tasks [26, 27]. Many use a generative model to construct the stored representation. Mem0 extracts and updates facts [3], Zep constructs a temporal knowledge graph [31], and Letta's agents edit memory files [17, 26]. Workflow memories similarly use a model to derive guidance from past agent runs [7, 25, 39].

Generative writing introduces three possible costs. First, extraction and update calls add ingestion cost as the history grows, even when no later question uses the stored information [3, 31]. Second, sampling can produce different stores from the same history, so reproducing a memory requires more than its construction code. Third, the writer can omit or alter details that later questions need. One audit of a Mem0 deployment classified 97.8 percent of 10,134 stored entries as junk [22]; this is evidence from one deployment rather than

a general failure rate. An agent that runs across thousands of sessions pays these costs on every one of them, while a benchmark score is paid for once, so we treat a memory result as a claim about accuracy together with tokens read, search time and build cost. We investigate what can be achieved without generative write calls. Our main LoCoMo comparison uses Mem0's published reader, judge and prompts on both sides.

Our memory stores each conversation turn verbatim, together with its speaker and date (Figure 2). To retrieve relevant evidence, we combine a keyword search for matching words with an embedding search for similarity in meaning. On LoCoMo, this produces an initial ranking of turns that we refine in three further stages. A dialogue rule passes part of each turn's score to the next turn in the same session, helping surface replies that do not repeat the question's words. A second rule boosts turns spoken by the person the question is about. Two local models, called cross-encoders, then rescore the leading candidates by evaluating each turn together with the question. A reader language model answers from the highest-ranked original turns, presented in chronological order. On LongMemEval-S, the keyword and embedding searches rank whole sessions without these later stages. We also adapt this retrieval approach to rank past agent trajectories on Proced-Mem [12].

The answer comparison on LoCoMo has similar aggregate accuracy under a shared reader, judge and prompts from Mem0's published evaluation [21]. On the 607 questions both systems have answered, our store scores 90.6 against 90.3 while reading 3,294 tokens per question against 6,956, with no generative write calls. This aggregate includes 385 development questions used to tune the search. On the 222 test questions alone, our store scores 82.0 against 85.6. The full test split was not completed, so the aggregate result does not establish parity on unseen questions.

Retrieval can also be measured without a reader. Our store finds every gold session in the top 10 for 97.4 percent of 470 LongMemEval-S questions; EdgeMem and MemGAS report 93.8 and 94.5, respectively [6, 42]. Our result includes scored questions from the 150-question development split used for tuning, and the comparison interprets the published retrieval metric as all-gold recall. On LoCoMo, the store retrieves 80.3 percent of gold turns in the top 10, against 64.0 for a flat fusion of the same two signals.

The error analysis identifies different weaknesses: model-written facts can lose detail on single-hop questions, while retrieval can omit a turn needed for a multi-hop answer. Two search changes motivated by these errors raise test all-gold recall from 90.2 to 94.4 at 150 turns. This measures the share of questions whose gold turns are all retrieved; no reader has been evaluated on the changed ranking. On Proced-Mem, the retriever reaches 0.957 mean average precision against the benchmark's 0.796. That retriever was selected on the same 40 queries used to score it.

Every main comparison in the paper is in Table 1 with its interval, and Figure 1 draws the main ones, so that a reader can see at once where the store beats, ties and loses. We mark the rows where tuning touched the scored questions, we state the judge on every row, and we claim a lead only where the difference lies outside its interval under a shared setup. Every run was registered with a prediction before it started, each miss is reported, and the runs that the model budget stopped are reported on the questions they finished and listed with their predictions in Section 5.7.

Six terms recur. The reader is the language model that answers from the retrieved text, and the judge is the language model that marks an answer right or wrong. A gold turn is a turn the benchmark marks as evidence for the answer, and all-gold recall at k is the share of questions whose gold turns are all among the top k retrieved. A tuned mark (‡) flags a row that includes questions the system was tuned on. A band is a comparison whose two sides were judged by different models; it places a system near another and does not rank them.

2 Related Work

We organize prior work by its stored representation: model-written summaries, facts or notes; verbatim interactions; and guidance derived from past agent runs. This separates the choice of what to store from the method used to retrieve it.

Model-written stores. MemoryBank and ReadAgent use a model to write summaries or gists of past sessions [15, 49]. Mem0 extracts candidate facts from each exchange with one model call and reconciles them against stored facts with another; its graph variant also extracts entities and relations [3]. Zep builds a temporal

Table 1: Every main comparison in one place. Beats and loses mean the difference lies outside its 95% interval or margin. Tie means it does not. Band means the two rows differ in judge and cannot be ranked. ‡ marks a row that includes questions the system was tuned on. One judge means both systems' existing answers were re-scored by Claude Sonnet 5.5 under the same prompt. Section 4 gives the interval methods and the caveats on the published rows.

Benchmark and measure	Ours	Comparator	Difference	Setup	Verdict
LongMemEval-S, all gold sessions in top 10, 470 q.‡	97.4	EdgeMem 93.8	+3.6, margin 2.6	no reader; published metric inferred	beats
LongMemEval-S, all gold sessions in top 3, 470 q.‡	84.0	EdgeMem 81.5	+2.6, margin 4.8	no reader; published metric inferred	tie
LoCoMo, share of gold turns in top 10, 1,152 test q.	80.3	flat fusion 64.0, our own baseline	+16.3 [+13.8, +19.2]	no reader	beats
Proced-Mem, MAP, 40 queries‡	0.957	MiniLM baseline 0.796	+0.16; judge moves a ranking by up to 0.08	same judge; no held-out split	beats
LoCoMo single-hop accuracy, 256 shared q.‡	95.7	Mem0 90.6	+5.1 [+2.6, +9.2]; the 56 test q. alone +7.1 [−1.8, +16.1]	same reader, judge, prompts	beats; tie on test alone
LoCoMo accuracy, 607 shared q.‡; tokens 3,294 vs 6,956	90.6	Mem0 90.3	+0.3 [−3.6, +3.6]	same reader, judge, prompts	tie
LoCoMo accuracy, the 222 test q. both answered	82.0	Mem0 85.6	−3.6 [−7.1, −0.5]	same reader, judge, prompts	loses
LoCoMo accuracy, 607 shared q.‡, one judge	90.1	Mem0 90.0	+0.2 [−4.6, +3.5]	same reader and prompts; Claude Sonnet 5.5 judges both	tie
LoCoMo accuracy, the 222 test q., one judge	80.6	Mem0 85.6	−5.0 [−8.1, −1.4]	same reader and prompts; Claude Sonnet 5.5 judges both	loses
LoCoMo multi-hop accuracy, 130 shared q.‡	88.5	Mem0 94.6	−6.2 [−11.3, −2.5]	same reader, judge, prompts	loses
LongMemEval-S accuracy, 319 test q.‡; tokens 19,125 vs 6,787	94.0	Mem0 93.4 on 500	+0.6	same reader	band
BEAM 1M pass rate, 652 q.	71.8	Mem0 70.1 on 700	+1.7	same reader	band
BEAM 1M pass rate, 652 q., one judge	61.3	Mem0 63.9 on 700	−2.6; intervals [57.6, 65.0] and [60.2, 67.3]	same reader; Claude Sonnet 5.5 judges both	tie
LoCoMo search time per question, seconds	1.05	Mem0 current 3.57, hosted mean; Mem0 2025 paper 0.200, own hardware	−2.52 s; +0.85 s	our laptop median; no shared hardware	beats the hosted mean; loses to the 2025 row
LoCoMo accuracy, best published row, any reader‡	90.6	Zep 94.7, gpt-5.4, third party	−4.1	different reader and judge	band
LongMemEval-S accuracy, best published row, any reader	94.0	Mastra 94.9, gpt-5-mini	−0.9	different reader and judge	band

knowledge graph with separate model passes for entities, facts, dates and contradictions [31]. Supermemory and Hindsight also extract facts at ingest [14, 35]. Other systems give agents control over memory files or notes. MemGPT lets the agent move facts between its context window and an external store [26], and Letta adds a background agent that rewrites memory files [17]. A-MEM writes linked notes and revises earlier ones [43], while MemGAS stores several granularities, including model-written summaries [42]. These systems incur generative calls when constructing at least part of their memory representation.

Verbatim stores. EdgeMem and Chain-of-Memory are the closest comparisons to our approach because they retain turns and search them with local models. EdgeMem annotates verbatim turns with spaCy and searches two keyword indices, without generative model calls before the reader. It reports retrieval on LongMemEval-S [6]. Chain-of-Memory builds chains of turn embeddings using a large embedding model and, as far as its method section states, generates no text at build or search time [44]. Our store adds an embedding search and two cross-encoder rerankers relative to EdgeMem, and a keyword search and two dialogue rules relative to Chain-of-Memory. Related retrieval work defers or simplifies graph construction, as in LazyGraphRAG and

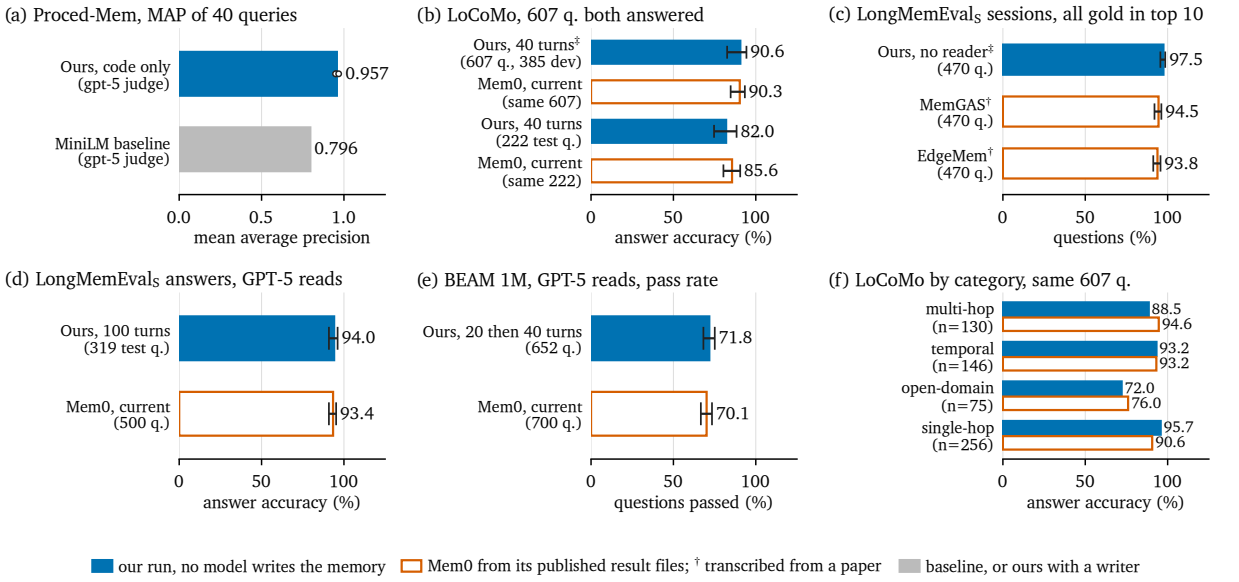


Figure 1: The main comparisons. Filled bars are our runs. Hollow bars are other systems, read from their published per-question result files unless marked † (transcribed from a paper). Whiskers are 95% intervals: conversation-cluster bootstrap in (b), Wilson score intervals for counts out of n elsewhere. (b) and (f) use the 607 LoCoMo questions that both our GPT-5 run and Mem0's file have answered, with Mem0's answer and judge prompts and GPT-5 as reader and judge on both sides; 385 of the 607 are development questions on which the search weights were set (‡), and the lower pair in (b) is the 222 test questions alone. (c) is retrieval with no reader, read as $\text{Recall}_{\text{all}}$@10. (d) Our row uses the official LongMemEval judge; Mem0's row uses GPT-5 with its own judge prompt. (e) Our row uses BEAM's judge on gpt-4.1-mini; Mem0's uses GPT-5 with its own prompt. Dots in (a) are the three judge passes.

LinearRAG [23, 50]. LongMemEval also shows that indexing at the right granularity recovers much of the gap between full context and an oracle history containing the answer evidence [40].

Procedural memories. These systems store information about how tasks were carried out. Agent Workflow Memory uses a model to induce workflows from past trajectories; its appendix also reports a rule-based variant scoring 35.6 on WebArena against 35.5 for the model-written variant [39]. Memp stores the raw trajectory, a model-distilled script, or both [7]. ReasoningBank uses a model to judge success and extract up to three items per trajectory [25]. ExpeL, Reflexion, Synapse, Voyager and SkillWeaver write insights, exemplars or executable skills with a model [34, 37, 46–48]. We evaluate trajectory retrieval on Proced-Mem, which has 40 queries over a bank of agent trajectories, model-judged relevance and no held-out split [12].

Benchmark and evaluation limits. LoCoMo has ten two-person conversations and 1,986 questions. Vendors score the 1,540 questions outside the adversarial category with the judge Mem0 introduced [3, 19]. Two audits found 6.4 and 7.0 percent of its answer keys wrong, and one found that the judge accepted 62.81 percent of deliberately wrong on-topic answers [4, 28]. Scores also depend on the reader and evaluation setup: full context scores 72.90 with gpt-4o-mini in Mem0's table and 92.85 with gpt-5-mini in a later study [29]. Such cross-reader comparisons do not isolate the memory's contribution. LongMemEval has 500 questions over histories of about 115k tokens. Its September 2025 release replaced 454 histories, so results depend on which release is used [4, 40]. BEAM has conversations of 128k to 10M tokens with 20 questions per conversation over ten abilities. Questions are graded item by item against rubric lists, but published rows do not always specify the judge model and prompt [36]. We use a shared reader and judge wherever a published per-question file permits it. Otherwise, we state the judges and treat differences as comparisons across evaluation setups rather than as a ranking of memories.

Table 2: Methods and models used for retrieval and evaluation. The retrieval components do not generate text. The LLM judge in the last row scores answers in the experiments and is not part of the store.

Method or model	What it does	Where	Settings
Stemmed BM25 [18, 33]	Scores a turn by the rare words it shares with the question, after lowercasing, stopword removal and Snowball stemming	Stage 1; LongMemEval-S sessions	$k_1 = 1.5$, $b = 0.75$
Sentence embedder, bge-base-en-v1.5 [41]	Maps the question and each turn chunk to a vector; the dense score is their cosine	Stage 1	450-token chunks, 50 overlap, 512-token truncation
bge-small-en-v1.5; bge-m3 [2]	The same role with a smaller and a larger embedder	Retrievers 5a and 5r; Proced-Mem	
z-score fusion	Puts the two score lists on one scale and adds them with fixed weights	Stage 1	0.25 BM25, 0.75 dense
Reciprocal rank fusion [5]	Adds $1/(k + \mathrm{rank})$ for each item over two ranked lists	5a, 5r, Proced-Mem	$k = 20$ on Proced-Mem
Forward contiguity	Passes half of a turn's positive score to the next turn in its session	Stage 2	weight 0.5
Speaker prior	Lifts every turn said by the person the question is about	Stage 3	2 standard deviations
spaCy `en_core_web_sm` [9]	Parses the question to find its subject or possessor; named entities for the graph ablation	Stage 3; entity graph	
Cross-encoders ms-marco-MiniLM-L-6-v2 and bge-reranker-v2-m3 [2, 38]	Read the question and a turn together and output one relevance score	Stage 4	top 100; weights 0.2 and 0.4
bge-reranker-base	The same role with one model	5r; LongMemEval-S budget rule	top 100; top 150
Personalized PageRank over a spaCy entity graph [8]	Walks from the question's entities to the turns that mention them	Ablation only	damping 0.5, weight 0.1
IDF-weighted token overlap	Counts rare shared stemmed words between a query and a trajectory	Proced-Mem	at least two shared tokens
LLM judge	Scores an answer against the gold answer	Evaluation only	Mem0's and the benchmarks' prompts

3 Methods

We study Snapgraph as an agent memory. A companion paper measures its retrieval [13]; here, reader models answer questions from the original text that it retrieves. This lets us compare a verbatim store with model-written memories. The store holds one kind of unit at a time, and we use it on two: conversation turns, for the three chat benchmarks that ask a memory to remember facts people said, and whole agent trajectories, for Proced-Mem, which asks it to remember how a task was done. The write path and the fusion stage are the same for both, and the dialogue stages apply only to turns. The write path builds two indexes without generative model calls. The LoCoMo read path ranks turns in four stages, each defined below. Table 2 lists the methods and models, their roles and their settings.

3.1 Problem

The retrieval task is to rank the history items that contain the answer above those that do not. A history H is a sequence of sessions $S_1, \ldots, S_m$. Each session has a date and an ordered list of turns; a turn $t_i = (\mathrm{speaker}_i, x_i)$ contains a speaker and text. Given a question q and its date, the store returns a strict ranking of turns, or of sessions on LongMemEval-S. A retrieval budget k selects the first k items. The reader receives the selected text unchanged, together with its speaker and date, and generates an answer.

Local embedding and reranking models produce vectors and relevance scores rather than stored prose. At retrieval time, the system receives the history and question, without benchmark answer labels or question-type labels. Labels are used to score and select settings on the splits described in Section 4; the Proced-Mem selection is described in Section 3.6. We evaluate retrieval by the ranks of the gold items, which the benchmark marks as containing answer evidence. We evaluate the combined memory and reader by answer accuracy.

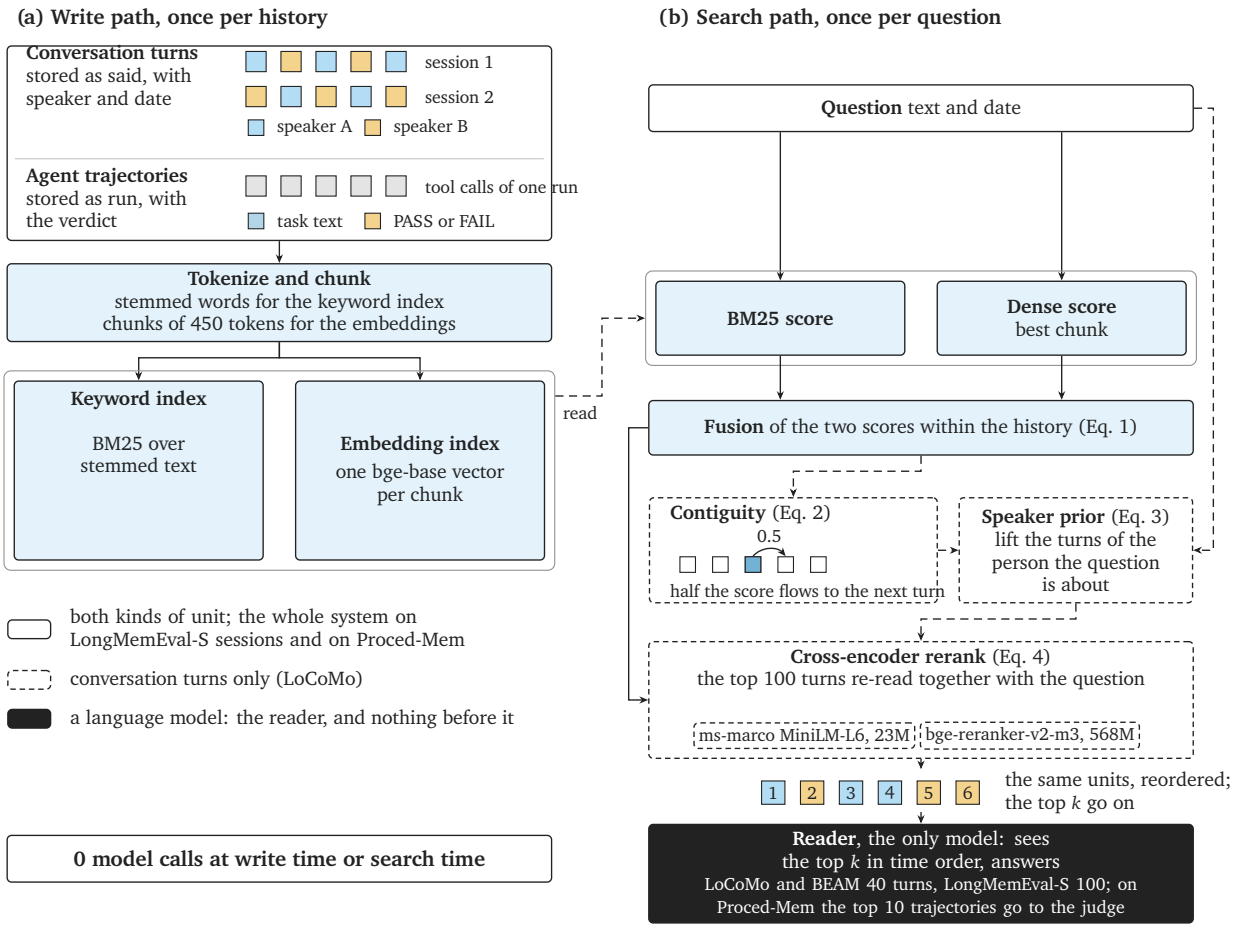


Figure 2: The store. (a) The write path runs once per history and keeps two kinds of unit as they were, conversation turns with their speaker and date or whole agent trajectories with their verdict, then builds a keyword index and an embedding index over them; no model writes anything. (b) The search path runs once per question: the two scores are fused (Equation 1), contiguity and the speaker prior reorder the turns (Equations 2 and 3), two local cross-encoders rescore the top 100 (Equation 4), and the same units come out reordered. Solid boxes serve both kinds of unit and are the whole system on LongMemEval-S sessions and Proced-Mem; dashed boxes serve conversation turns only. The reader, the only language model, sees the top k in time order and answers.

3.2 Storage

The write path builds two indexes using local compute, without generative write calls. Turns longer than 450 tokens are cut into chunks of 450 tokens with 50 tokens of overlap; shorter turns form one chunk. Each chunk is embedded with bge-base-en-v1.5 [41], with inputs truncated at 512 tokens and vectors L2-normalized. The same turns are indexed by BM25 [33] in the Lucene variant of bm25s [18], with $k_1 = 1.5$ and $b = 0.75$. Its input is lowercased word tokens with English stopwords removed and Snowball stemming [30]. On LoCoMo, each turn's BM25 text is prefixed with its speaker's name so that keyword search can distinguish the speakers. Embedding the 5,882 LoCoMo turns takes 36 seconds on a laptop CPU. Building a LongMemEval-S history takes 0.19 seconds per session (Section 6). With fixed index settings, the deterministic construction produces the same store from the same history.

3.3 Search

The LoCoMo read path has four stages. Each transforms the scores assigned to turns in one history, without adding or removing a turn. Its output is therefore a reordering of the same history. Removing a stage in an

ablation changes the ranking rather than the available content. The LongMemEval-S configuration uses only the first stage, as described under Configurations below.

Fusion. For a question q, every turn receives a BM25 score and a dense score, the cosine between the question embedding, which carries the bge retrieval instruction prefix, and the turn's best chunk. Each score vector is z-normalized over all turns of the history, $z(x) = (x - \bar{x})/\sigma_x$ with the population standard deviation, so that the two channels sit on one scale, and the two are summed with fixed weights,

$$s_i = 0.25\, z(\mathrm{bm25})_i + 0.75\, z(\mathrm{dense})_i. \tag{1}$$

The BM25 weight was chosen on the development split from $\{0.25, 0.4, 0.5\}$, and the flat fusion that every ablation starts from is the same sum at weights 0.4 and 0.6, fixed before tuning. The dense channel carries most of the weight because it matches paraphrase, and the keyword channel is kept because it catches the rare word, a name or a place, that an embedding smooths over.

Order. In two-speaker dialogue the turn that answers a question is usually the reply to the turn whose wording the question repeats, and therefore the turn that scores well is often the one just before the turn the reader needs. Each turn passes half of its positive score to the next turn in the same session,

$$s'_i = s_i + 0.5\ \max(s_{i-1}, 0), \tag{2}$$

applied only when t_{i-1} and t_i are in the same session. The weight 0.5 was tuned on the development split, and the same forward asymmetry appears in human free recall [10, 11]. On LongMemEval-S this stage is skipped, because the evidence there is usually one user turn inside a long session and a neighbour takes a slot that holds no evidence.

Speaker. A spaCy parse of the question [9] finds the speaker it is about by taking the single speaker named or, when both are named, the grammatical subject and failing that the possessor, so that no speaker is chosen when the two names are coordinated. Every turn said by that speaker gains two standard deviations of the current scores,

$$s''_i = s'_i + 2\,\mathrm{std}(s') \cdot \mathbb{1}[\mathrm{speaker}_i = \mathrm{person}(q)], \tag{3}$$

so that a question about Caroline's plans is answered from Caroline's turns even when her friend's turns use the same words.

Rerank. Two local cross-encoders, each of which reads the question and a turn together and outputs one relevance score, rescore the top $N = 100$ turns under s'', since they are too slow to run over a whole history but cheap over a hundred candidates. They are ms-marco-MiniLM-L-6-v2 with 23M parameters [1, 24, 38] and bge-reranker-v2-m3 with 568M [2], both run through sentence-transformers [32] with inputs truncated at 512 tokens, and within the N candidates

$$r_i = 0.4\, z(s'')_i + 0.2\, z(\mathrm{MiniLM})_i + 0.4\, z(\text{bge-v2-m3})_i, \tag{4}$$

with the z-scores taken over the candidates and the reranked turns kept above all others. The output is the ranked list the reader cuts.

Configurations. The graph-free ranker is the four-stage LoCoMo method defined above. It is used in the main LoCoMo answer comparisons and the retrieval rows bearing that name. On LongMemEval-S, the unit is instead a session: the BM25 document concatenates the session's text, and the dense score is the maximum over its chunks. Stage 1, with BM25 weight 0.25, is the whole session-ranking system.

The ranker with entity graph is the earlier frozen LoCoMo configuration used for the graph ablation. It adds a spaCy entity graph searched with personalized PageRank, an entity field in the BM25 text and backward contiguity at weight 0.25. Together, these additions contribute 0.9 to 1.9 points of turn recall in Section 7. The graph-free ranker omits them. Secondary answer rows report both configurations.

Table 3: Rank of the gold turn after each stage for three development questions, each with one gold turn, in a 419-turn LoCoMo conversation. S1 is Equation 1; the BM25 and dense columns are its two inputs ranked alone. S2 to S4 are Equations 2 to 4, with names kept out of S1 and 300 candidates in S4.

Question	BM25	Dense	S1	S2	S3	S4
What are Caroline's plans for the summer?	111	313	320	11	6	6
How does Melanie prioritize self-care?	105	122	144	5	4	4
How long has Melanie been practicing art?	367	3	12	3	2	2

Later search changes. After the head-to-head analysis in Section 5.2, two further changes were chosen on the development split. The first removes the named speakers from the question seen by Stage 1; Stages 3 and 4 still receive the full question. The second increases the Stage 4 candidate set to $N = 300$. Section 5.3 reports their retrieval results on both splits. No answer row uses these later changes: the reader evaluations use configurations frozen before them.

Earlier retrievers. The secondary rows in Section 5.2 also include 5a and 5r. Retriever 5a uses bge-small-en-v1.5, cleans and stems the question, and combines keyword and embedding rankings by reciprocal rank fusion. This adds $1/(k + \mathrm{rank})$ for each turn across the two lists [5]. Retriever 5r reranks the top 100 turns of 5a with bge-reranker-base and fuses that ranking back by the same rule. Both were chosen on all LoCoMo questions before the split existed and carry the ‡ mark.

3.4 Example

One stage, contiguity, does most of the work on questions whose answer shares no words with the question, and Table 3 shows this by following three development questions from a 419-turn conversation, each with one gold turn, and giving the gold turn's rank after each stage. For "What are Caroline's plans for the summer?" the gold turn is Caroline's reply, which repeats none of the question's words, and therefore BM25 ranks it 111th and the embedder 313th; the turn before it, the question that prompted the reply, ranks well, and Stage 2 passes it half of that score, which lifts the reply to 11th, after which the speaker prior lifts it to 6th and the rerank leaves it there. The second question follows the same path, since its gold turn is also a reply that repeats none of the question's words and the contiguity stage lifts it from 144th to 5th, while for the third the embedder alone ranks the gold turn 3rd and the later stages keep it near the top. These three show the mechanism of Stage 2; across the whole test split the two cross-encoders are the largest single contributor to recall at 10 and contiguity the second (Section 7).

3.5 Reader

For the like-for-like comparisons, we hold the reader prompt and input layout to Mem0's current format: one line per retrieved turn, with its date. The turns appear in time order, with their speaker and original text. The LoCoMo context-budget comparison uses three settings. The fixed cut provides the top 40 turns. The adaptive cut first provides the top 10, with an added instruction allowing the reader to answer NOT FOUND; if it does, the reader receives the top 40 and answers again. The full-context run provides every turn of the conversation. The registered, unrun budget-matched LongMemEval-S arm would provide every user turn in the top 60 and an assistant reply only when it ranks in the top 10. This rule reflects the evidence usually appearing in a user turn and the greater length of assistant replies: 358 tokens on average against 53 for user turns.

3.6 Procedures

We adapt verbatim storage and retrieval to past agent runs on Proced-Mem [12]. The retrieval unit is a complete trajectory rather than a conversation turn, and the query describes a task. The retriever embeds

task text with bge-m3 [2]. In place of BM25, it uses IDF-weighted token overlap, which gives rare shared stemmed words more weight and requires at least two shared tokens. Reciprocal rank fusion combines the lexical and embedding rankings with $k = 20$, weighting the embedding ranking two to one. The system keeps each trajectory whole and writes no generative summary of it.

This configuration was chosen by an offline sweep of 427 rule-built variants on the benchmark's own relevance labels. Section 5.6 reports the sweep and the limitation of selecting and scoring the retriever on the same queries.

4 Setup

We evaluate conversational recall on LoCoMo, LongMemEval-S and BEAM, and procedural retrieval on Proced-Mem. The reader generates an answer from retrieved evidence; the judge grades that answer. Our main LoCoMo comparison with Mem0 shares the reader, judge and both prompts. The LongMemEval-S and BEAM comparisons share the GPT-5 reader but use different judges, so their answer scores do not establish a ranking between the systems. We specify the data splits, input budgets and evaluation conditions below.

LoCoMo. LoCoMo contains ten two-person conversations [19]. We score the 1,540 questions outside the adversarial category. Conversations 26, 30 and 41 form the development set of 385 questions; the other seven form the test set of 1,155. Every test accuracy therefore refers to these 1,155 questions or an explicitly stated subset. The matched comparison uses GPT-5 as both reader and judge, with Mem0's current answer and judge prompts copied from its evaluation repository [21]. Mem0's scores come from the per-question results in that repository. Secondary evaluations use Mem0's 2025 judge prompt on gpt-4o-mini [3] and the reader named in each table. They provide context rather than a matched comparison with Mem0's current release.

Retrieval evaluations require usable evidence labels. The ranker study includes 383 development and 1,152 test questions; the earlier study of retrievers 5a and 5r includes 383 and 1,153. All-gold recall at k is the share of questions for which every gold turn is among the top k retrieved turns. A gold turn is one the benchmark labels as evidence for the answer.

LongMemEval-S. We use the cleaned release of 500 questions [40]. Each question has a history of about 115k tokens across 48 sessions. Question ids are sorted by SHA-256 to define the splits.

Answer evaluations use the first 100 questions for development and the remaining 400 for test. GPT-5 reads the top 100 turns under Mem0's current answer prompt. We grade answers with the official judge, gpt-4o-2024-08-06, using its per-type prompts. Mem0's published results instead use GPT-5 as judge with its own prompt. The reader is shared, but the judge is not. A secondary evaluation uses Claude Sonnet 5.5 over 40 turns and the official prompt on all 500 questions.

The ranker retrieval study uses a separate split: 150 development questions and 350 test questions. Excluding abstention questions leaves 331 scored test questions. The earlier 5a/5r retrieval study used the 100/400 split, with 375 scored test questions (Appendix B). Session-level $\text{Recall}_{\text{all}}$@k is the share of questions for which every gold session is among the top k retrieved sessions.

BEAM. BEAM contains conversations of 128k to 10M tokens, with 20 questions per conversation covering ten abilities [36]. We use the 1M-token tier: 35 conversations and 700 questions. GPT-5 first reads the top 20 turns and receives the top 40 if it answers NOT FOUND. BEAM's judge prompt on gpt-4.1-mini scores each rubric item as 0, 0.5 or 1. A question passes when its mean item score is at least 0.5, following Mem0's rule. Mem0's published answers are judged by GPT-5 with its own rubric prompt. No configuration was selected on BEAM; our results use a single run of the frozen configuration.

Proced-Mem. Proced-Mem has 40 queries over a bank of agent trajectories [12]. Each query returns ten trajectories. A gpt-5 judge at low effort scores each trajectory from 1 to 10, with scores of at least 6 counted as relevant. We use the repository's runner unmodified and report the mean of three judge passes because a single pass can move a fixed ranking by up to 0.08. Configuration selection and evaluation use the same 40 queries; the benchmark has no held-out split.

Protocol. Before each run, we registered its model, settings, memory-file hash and numbered predictions with ranges in a protocol file. Scripts read the result files and mark each prediction as held or missed. Appendix A lists the registrations and outcomes.

For the main conversation ranker, settings were selected on development data before the frozen test evaluation. Retrieval scores were reproduced on reruns from an empty cache. The two later search changes in Section 5.3 were also selected using development labels, then ranked once on the test split with their settings frozen. Earlier retrievers selected on scored questions are marked as tuned below. Budget-stopped runs are reported only on completed questions; unstarted runs remain listed with their predictions in Section 5.7.

Statistics. Unless stated otherwise, bracketed ranges are 95 percent intervals and differences in accuracy or recall are percentage points. We use Wilson score intervals for proportions of yes-or-no outcomes. LoCoMo accuracy comparisons use 2,000 bootstrap draws that resample whole conversations, preserving dependence among questions from the same conversation. Ranker comparisons use paired resampling so both systems are scored on the same sampled questions; captions specify where the resampling unit is a conversation rather than a question. Comparisons with published retrieval results use the independent-row margins stated in Section 5.1.

Published and tuned results. Transcribed results carry the mark † and receive an interval only when their underlying count is known. The mark ‡ identifies results whose retriever was selected or tuned on questions also included in the reported score. This applies to the earlier 5a/5r LoCoMo retrievers, selected before a split existed; the LongMemEval-S results on all 470 questions, including scored questions from the 150-question development split used for tuning; and the pooled 607-question LoCoMo comparison, including 385 development questions. Proced-Mem also carries this qualification because selection and evaluation share the same queries. In Table 1, intervals are conversation-cluster bootstraps on LoCoMo and Wald margins between independent rows on LongMemEval-S retrieval. The published LongMemEval-S retrieval values are read as $\text{Recall}_{\text{all}}$@k; that reading is inferred from the published tables. The two best-published accuracy rows carry a different reader and judge from ours. The search-time row sets our laptop median against two published figures: the mean latency Mem0's result file logs on its hosted service, which includes network time, and the 95th-percentile time Mem0's 2025 paper reports on its own hardware for an earlier release. Our 1.05 s is below the first and above the second. Tokens in that table are the reader's input per question.

5 Results

We first measure retrieval without a reader, then answer accuracy using the retrieved evidence. On LoCoMo, the store and Mem0 have similar accuracy on the shared set at about half the reader input, but most questions in that set were used for development. The store is less accurate on the partial test comparison, particularly on multi-hop questions. On LongMemEval-S and BEAM, the observed accuracy gaps are within two points, but the judges differ and the results do not rank the two systems.

5.1 Retrieval

On session-level retrieval over all 470 non-abstention LongMemEval-S questions, the store finds every gold session in the top 10 for 97.4 percent, against 93.8 for EdgeMem and 94.5 for MemGAS, the strongest published results in Table 4 (Figure 3). At the top 3, it scores 84.0 against 81.5 and 78.5. This comparison includes questions used for tuning, and we infer that the published R values measure $\text{Recall}_{\text{all}}$@k from their table. Supermemory reports 97 on LongMemEval-S under a label of Recall@20 with aggregation [35]. That label's metric is not stated, so the row is not in Table 1.

On the 331 test questions, which exclude the questions used for tuning, the store scores 97.3 at 10 and 84.3 at 3. We do not compare these values directly with the published all-question results. The test subset is easier for at least one system: an EdgeMem-style reimplementation scores 81.0 $\text{Recall}_{\text{all}}$@3 on test and 79.2 on all 470. EdgeMem's per-question output is not public, so we cannot score it on the same test subset.

Against EdgeMem on all 470 questions, the gap at 3 is +2.6, inside the 95 percent Wald margin of ±4.83; the gap at 10 is +3.6, outside the margin of ±2.60. These margins treat the results as unrelated samples and

System	Generative writer	n	$\text{Recall}_{\text{all}}$@3	$\text{Recall}_{\text{all}}$@10	NDCG@10
All 470 non-abstention questions					
MPNet†	no	470	66.17	85.11	80.63
Contriever†	no	470	71.06	90.00	84.29
SeCom†	yes	470	71.06	89.15	85.11
HippoRAG 2†	yes	470	75.53	91.28	88.73
MemGAS†	yes	470	78.51	94.47	89.96
EdgeMem†	no	470	81.49	93.83	92.80
Flat fusion‡	no	470	83.62	96.60	94.95
Ours‡	no	470	84.04	97.45	94.95
Test split					
Flat fusion	no	331	84.3	96.7	95.5
Ours	no	331	84.3	97.3	95.3

Table 4: Session-level retrieval on LongMemEval-S. Flat fusion is stemmed BM25 at weight 0.4 plus bge-base at 0.6, fixed before tuning; ours retunes the weight to 0.25. † Transcribed: EdgeMem from its paper, the others as EdgeMem transcribes them from the MemGAS paper. ‡ The 470 scored questions include questions from the 150-question development split used for tuning. NDCG@10 uses binary relevance, read as in the published table.

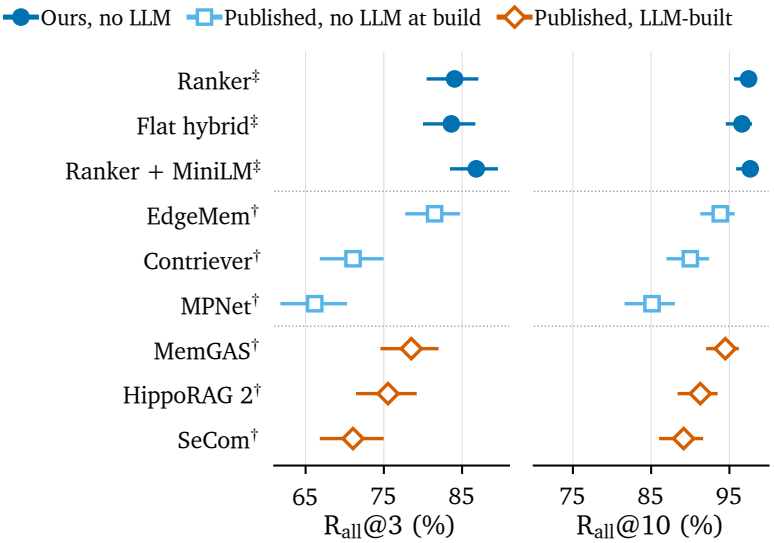


Figure 3: LongMemEval-S session $\text{Recall}_{\text{all}}$@3 and $\text{Recall}_{\text{all}}$@10 on 470 non-abstention questions, with 95% Wilson intervals. ‡ Our systems on all 470 scored questions, including questions from the 150-question development split used for tuning. Published values are read as $\text{Recall}_{\text{all}}$@k, an inferred metric; rows other than EdgeMem come from the MemGAS paper through EdgeMem.

do not account for tuning on the 150-question development set. The metric also imposes a ceiling at small k: of the 52 test misses at 3, 32 have three or more gold sessions, and 24 have more gold sessions than the top 3 has slots.

On LoCoMo no published memory reports turn recall, so we compare against our own flat fusion of the same two signals. The graph-free ranker used in the main answer comparison finds 80.3 percent of gold turns in the top 10 on the seven test conversations, against 64.0 for flat fusion and 54.8 for stemmed BM25 alone (Table 10, Figure 10b). It finds 88.5 percent at the top 50. The gain over flat fusion at 10 is 16.3 [+13.8, +19.2] points. The frozen configuration with the entity graph finds 81.5 at 10 and 90.1 at 50; removing its two cross-encoders costs 6.1 of its 17.5 points of gain over flat fusion (Section 7).

Table 5: LoCoMo under Mem0's current setup: GPT-5 reads, GPT-5 judges, Mem0's answer and judge prompts. Accuracy by category and overall, with n in the header or row label. Mem0's rows are scored from its published per-question file on our split. Our rows read the top 40 turns of the graph-free ranker. The matched test comparison is 82.0 against 85.6 on the same 222 questions; Mem0's full-test result uses all 1,155. The 222-question sample covers the four categories almost equally. Weighting our category accuracies by the full test split's counts gives 88.7, a separate estimate. Recall is the share of questions with every gold turn among the turns read. Tokens are the reader's input per question. The test run stopped at 222 questions when the budget ran out. ‡ The 607 include the 385 development questions on which the ranker's weights were set. Brackets are conversation-cluster bootstrap intervals.

System and questions	Reader, judge	Multi-hop	Temporal	Open	Single	All	Recall	Tokens
Development, 3 conversations, 385 questions (74 / 90 / 21 / 200)								
Ours, 40 turns	GPT-5, GPT-5	93.2	97.8	90.5	**96.0**	**95.6**	86.8	3,345
Ours, 100 turns (200 of 385)	GPT-5, GPT-5	94.3	100.0	90.5	95.2	96.0	–	4,202
Mem0, current	GPT-5, GPT-5	**95.9**	93.3	**95.2**	91.5	93.0	–	6,956
Test, 7 conversations, 1,155 questions (208 / 231 / 75 / 641)								
Ours, 40 turns, 222 done	GPT-5, GPT-5	82.1	85.7	64.8	94.6	82.0	69.8	3,206
Mem0, same 222 questions	GPT-5, GPT-5	–	–	–	–	85.6	–	–
Mem0, all 1,155 questions	GPT-5, GPT-5	92.3	92.6	70.7	92.5	91.1	–	6,956
The 607 questions both have answered‡ (130 / 146 / 75 / 256)								
Ours, 40 turns	GPT-5, GPT-5	88.5	93.2	72.0	**95.7**	**90.6** [82.6, 94.3]	–	3,294
Mem0, current	GPT-5, GPT-5	**94.6**	93.2	**76.0**	90.6	90.3 [84.7, 93.4]	–	6,956

5.2 LoCoMo

The LoCoMo comparison uses GPT-5 as reader and judge, with Mem0's current answer and judge prompts on both sides. Our reader receives the top 40 turns. The run completed all 385 development questions and 222 of the 1,155 test questions before the budget ran out; the remaining test questions were not run (Section 5.7). The run drew questions round-robin across the 28 conversation-by-category groups of the test split, each group shuffled with a fixed seed, so the 222 finished questions cover the four categories almost equally. We therefore report the matched partial test result separately from the development and pooled results (Table 5, Figure 1b).

On the same 222 test questions, the store scores 82.0 against Mem0's 85.6. The paired difference is −3.6 [−7.1, −0.5] under the conversation-cluster bootstrap over seven conversations. We answer 8 questions that Mem0 misses and miss 16 that Mem0 answers (exact McNemar $p = 0.15$). This sample covers the four question categories almost equally, over-sampling the harder categories. Weighting our category accuracies by their counts in the full test split gives an estimate of 88.7. That estimate is distinct from the matched 222-question comparison above.

On the 385 development questions used to set the ranker's weights, the store scores 95.6 against Mem0's 93.0, a difference of +2.6 [−2.0, +8.6]. Pooling development with the completed test questions gives 90.6 against 90.3 on 607 shared questions. The store reads about 3,300 tokens per question, against Mem0's 6,956. Because 385 of these 607 questions are from development, the pooled comparison carries a tuned mark in Table 1 and does not establish a tie on held-out data.

Mem0's scores come from its published per-question file [21], which lets us score the same subsets. In that release, GPT-5 answers from the top 200 memories and GPT-5 judges the answers. The file gives 91.6 overall on 1,540 questions, compared with 92.5 on Mem0's site; we use the file throughout. On our split, it gives 93.0 on the three development conversations and 91.1 on all seven test conversations.

The pooled result hides different error patterns. On the 607 shared questions, 385 of them from development, both systems answer 521, neither answers 30, ours alone answers 29 and Mem0 alone answers 27 (Figure 1f). The overall difference is +0.3 [−3.6, +3.6], with exact McNemar $p = 0.89$ for the 29 versus 27 discordant questions. On the 256 single-hop questions in this set, the store scores 95.7 against Mem0's 90.6, a difference of +5.1 [+2.6, +9.2]. We answer 18 that Mem0 misses and miss 5 that Mem0 answers ($p = 0.011$). On all 1,540 questions, 65 of Mem0's 130 errors are single-hop, close to the category's share of the questions (841 of 1,540), so the single-hop lead on the shared set comes from the discordant questions

rather than from a weak category. Examples include “Yoga” where the turn says lifting weights, and “keep believing in love” where the speaker said keep trying new things. These examples suggest a loss of detail during extraction; our reader instead receives the original turn.

On the 130 multi-hop questions, the store scores 88.5 against Mem0’s 94.6, a difference of −6.2 [−11.3, −2.5], with 2 versus 10 discordant questions ($p = 0.039$). The single-hop and multi-hop difference intervals both exclude zero. Temporal accuracy is level at 93.2; open-domain accuracy is 72.0 against 76.0 on 75 questions, with an interval that includes zero for the difference. On the 222 test questions alone, single-hop accuracy is 94.6 against 87.5, a difference of +7.1 [−1.8, +16.1] on 56 questions. Multi-hop accuracy is 82.1 against 92.9, a difference of −10.7 [−16.1, −5.4] on 56.

Of the 27 shared questions that Mem0 answers and we miss, 20 have at least one gold turn outside the 40 retrieved. For the other 7, every gold turn is present but the reader still answers incorrectly, for example by miscounting or choosing the wrong date. Among the 20 retrieval misses, the worst-ranked gold turn is within the top 60 for 2, the top 100 for 6, the top 150 for 8, the top 200 for 10 and the top 300 for 14. The remaining 6 require world knowledge not stated in the turn, such as that Mount Talkeetna is in Alaska, or have a label pointing to an unrelated turn. A writer can add such background knowledge to an extracted fact; a store retaining verbatim turns does not add it at ingest.

Even when retrieval supplies every gold turn, the reader sometimes answers incorrectly. In the test results, GPT-5 under Mem0’s prompt and judge answers 93.8 percent of multi-hop, 91.5 percent of temporal, 73.9 percent of open-domain and 96.2 percent of single-hop questions when every gold turn is among the 40 read. Weighting these conditional accuracies by the full test split’s category counts gives an estimated accuracy of about 93.4 under perfect retrieval. Mem0’s full-test score is 2.3 points below that estimate; our category-weighted estimate is about 4.7 points below it. These are distances from an estimated ceiling, not measured gains from improving search. The conversation-cluster intervals further limit the precision of comparisons on the 1,155-question test split.

One judge for both systems. The rows above use GPT-5 as judge, which is also Mem0’s choice, so we re-judged both systems’ existing answers with one other model, Claude Sonnet 5.5 under Mem0’s current judge prompt, registered beforehand with a prediction (Appendix A). On the 607 shared questions the store scores 90.1 against Mem0’s 90.0, a difference of +0.2 [−4.6, +3.5], with 29 questions we answer and Mem0 misses against 28 the other way (exact McNemar $p = 1.0$). On the 222 test questions it scores 80.6 against 85.6, a difference of −5.0 [−8.1, −1.4], with 6 against 17 discordant questions ($p = 0.03$), and on the 385 development questions 95.6 against 92.5, +3.1 [−1.3, +8.6]. The category pattern holds: +5.1 [+2.1, +8.6] on the 256 single-hop questions and −6.2 [−12.5, −2.7] on the 130 multi-hop questions. Across all 1,540 questions Mem0 scores 91.3 under this judge against 91.6 under its own. The two judges agree on 99.0 percent of Mem0’s answers, and on 99.0 percent of our development answers and 98.6 percent of our test answers. The tie, the test deficit and the category split are the same under either of the two model judges; neither has been checked against a human.

Other readers and judges. The secondary LoCoMo results in Table 6 and Figure 4 use other readers and Mem0’s 2025 judge prompt on gpt-4o-mini. They are not directly comparable with Mem0’s current release, because the reader or judge differs. These results include the earlier rank-fusion retriever (5a) and its reranked variant (5r), described in Section 3.

The choice of reader and judge materially changes accuracy. With Claude Sonnet 5.5 reading about 13 turns on average, the graph-free ranker scores 95.7 under the 2025 judge. Re-judging the same answers with GPT-5 and Mem0’s current prompt gives 91.3. With the same 40 turns from the rank-fusion retriever (5a), changing the reader from GPT-6 Sol to Sonnet raises accuracy from 91.1 to 94.2. These changes limit what can be inferred from results that use different readers or judges.

With Sonnet, the graph-free ranker’s adaptive reading policy scores 95.7 against 96.0 for reranked rank fusion (5r), while reading 13.2 turns on average instead of 40. The reader first receives the top 10 turns; if it answers NOT FOUND, it receives the top 40. This second pass ran for 163 of 1,540 questions. Those questions scored 74.2, against 98.3 for the rest. The adaptive policy reads 1,263 input tokens per question, against 3,191 for reranked rank fusion.

Table 6: Secondary LoCoMo rows: other readers and the 2025 judge (Mem0's 2025 judge prompt on gpt-4o-mini), on the 1,540 questions outside the adversarial category. Not like-for-like with Mem0's current release. Brackets are conversation-cluster bootstrap intervals. Test is the 1,155 scored questions on the seven test conversations. Recall is the share of questions with every gold turn among the turns read. ‡ Retriever chosen on all questions before the split. Costs are the answer model's tokens at list price and leave out the judge. † Transcribed rows.

Retriever and turns read	Reader	All 1,540	Test 1,155	Recall	Turns	Cost
BM25, 20 turns	GPT-6 Sol	82.3	–	59.7	20	$4.16
BM25, 40 turns	GPT-6 Sol	85.8 [84.0, 87.8]	84.7	64.8	40	$7.63
Rank fusion (5a), 40 turns‡	GPT-6 Sol	91.1 [90.0, 92.3]	90.4	75.3	40	$7.42
Full context	GPT-6 Sol	95.7 [94.5, 96.8]	95.5	all	601	$105.81
Rank fusion (5a), 40 turns‡	Sonnet 5.5	94.2 [92.9, 95.7]	93.1 [92.2, 94.2]	75.3	40	$10.51
Reranked rank fusion (5r), 40 turns	Sonnet 5.5	96.0 [94.7, 97.4]	94.9 [94.0, 95.9]	78.9	40	$10.80
Graph-free ranker, 10 then 40	Sonnet 5.5	95.7 [94.1, 97.5]	94.5 [93.2, 96.0]	76.4	13.2	$4.93
Ranker with entity graph, 10 then 40	Sonnet 5.5	95.8 [94.2, 97.5]	94.8 [93.3, 96.5]	77.9	12.9	$4.93
Rank fusion (5a), 40 turns‡	gpt-4o-mini	87.7	85.6	75.3	40	–
Graph-free ranker, 10 then 40	gpt-4o-mini	90.2	88.5	–	20.5	–
Mem0, 2025 paper†	gpt-4o-mini	66.88	–	–	–	–
Mem0 graph, 2025 paper†	gpt-4o-mini	68.44	–	–	–	–
Letta file search†	gpt-4o-mini	74.0	–	–	–	–
Zep, own re-run†	gpt-4o-mini	75.14	–	–	–	–
Full context†	gpt-4o-mini	72.90	–	–	–	–
Full context†	gpt-5-mini	92.85	–	–	–	–
Zep, 2026 report†	gpt-5.4	94.7	–	–	–	–

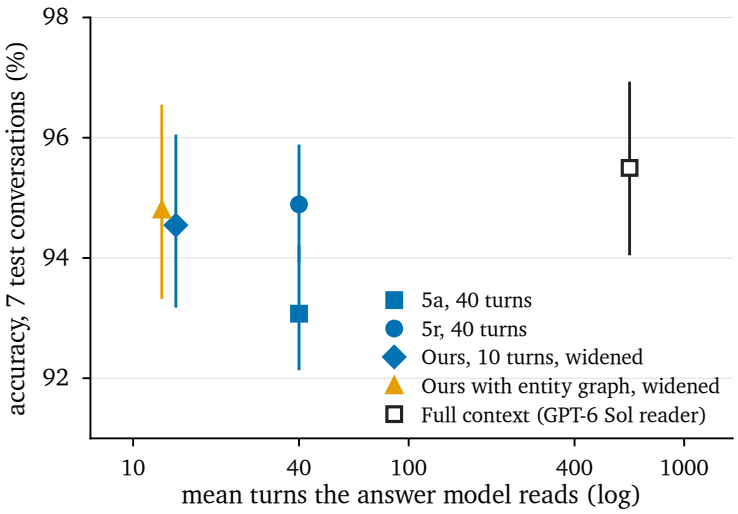


Figure 4: LoCoMo accuracy on the 1,155 scored questions of the seven test conversations, against the mean number of turns the reader sees, with 95% conversation-cluster bootstrap intervals, for the secondary rows. Full context uses a different reader.

With gpt-4o-mini as both reader and judge, the graph-free ranker with adaptive reading scores 90.2 at 1,457 tokens per question. Mem0's 2025 paper reports 66.9 for Mem0 and 68.4 for its graph variant, with 1,764 tokens per question for Mem0 [3]. That paper does not name its judge, so these are not matched comparisons. Other published results include Zep's own re-run at 75.14 and Letta's file-search agent at 74.0 with gpt-4o-mini [16, 45], full context at 92.85 with gpt-5-mini [29], and a third-party report of Zep at 94.7 with gpt-5.4 [4].

5.3 Search

Many of our misses have a gold turn ranked below the 40 turns given to the reader. We test two changes to the search: removing speaker names from the first-stage query and increasing the number of candidates rescored by the cross-encoders. Both are fixed rules chosen on the three development conversations, with labels used to score the settings. On the test split, ranked once with the chosen setting frozen, the changes together raise all-gold recall at 150 turns from 90.2 to 94.4 overall and from 77.4 to 86.1 on multi-hop questions (Figure 5). No reader has run on the revised ranking or its cascade variant below, so the gains are in retrieval recall; their effect on answer accuracy is unmeasured.

The first change keeps speaker names out of the first stage. A question such as "What things has Nate recommended to Joanna?" names both speakers. In the original ranker, those names reach BM25 and the embedder, giving high scores to turns in which the speakers greet each other by name. For that question, the top five turns included lines like "Nate: Keep going, Joanna!", while the recommendations ranked between 80 and 300. The speaker prior already uses the names to favour the right speaker. We therefore remove them from the question seen by the first stage, while the prior and cross-encoders retain the full question.

On development, this change raises all-gold recall at 40 turns from 87.2 to 88.8 overall and from 64.9 to 73.0 on the 74 multi-hop questions, recovering all evidence for six more questions. Temporal recall rises from 97.8 to 98.9, while single-hop recall falls from 93.0 to 92.5. Also removing names from the cross-encoder pairs gives only 68.9 on multi-hop, so we retain the names at that stage.

The second change gives both cross-encoders 300 candidates instead of 100. Of the 26 development multi-hop misses at 40 turns, 16 had a gold turn ranked past 100 and therefore outside the rerankers' reach. With speaker names kept, rescoring 300 candidates raises development recall at 40 from 87.2 to 88.3 overall and from 64.9 to 70.3 on multi-hop. Rescoring every turn gives 71.6 on multi-hop, suggesting that most of this gain is available at 300 candidates. Search takes 3.16 s per question instead of 1.05 s (Section 6), with no generative model tokens.

Combining the two changes gives the best overall development recall at 40 turns among the four settings tried: 89.6 overall, 71.6 on multi-hop, 98.9 on temporal, 63.2 on open-domain and 94.5 on single-hop. At 150 turns, development recall is 96.6 overall and 91.9 on multi-hop, against 93.7 and 82.4 for the original ranker. We also report this depth because grouping the dates of turns from the same session keeps the reader input below Mem0's token count (Section 6).

On test, the frozen combined setting gives recall at 40 turns of 86.3 overall and 62.0 on multi-hop, against 81.5 and 51.4 for the original ranker. At 150 turns, it gives 94.4 overall and 86.1 on multi-hop, against 90.2 and 77.4. Both results at 150 fall within the registered ranges: 92 to 94.5 overall and 83 to 89 on multi-hop. Recall at 150 is 99.1 for single-hop, 94.3 for temporal and 76.7 for open-domain questions.

A cascade reduces the extra search time: the 23M-parameter cross-encoder scores all 300 candidates, and the 568M-parameter cross-encoder scores only the top 100 of those. On development, its overall recall is 89.6 at 40 turns and 96.3 at 150, against 89.6 and 96.6 when both models score all 300 candidates. Its multi-hop recall at 40 is 73.0 against 71.6. Search takes 1.25 s per question. On the test split, ranked once with a registered prediction, the cascade gives overall recall of 86.1 at 40 and 93.8 at 150, against 86.3 and 94.4 for the full 300-candidate setting. Multi-hop recall is 61.1 and 85.1, against 62.0 and 86.1. The prediction at 40 held; the result at 150 missed its half-point band by 0.1 on the low side. The cascade therefore gives up about half a point of recall at 150 while taking 40 percent of the search time of the full 300-candidate setting.

Other development experiments did not improve recall. Searching again with the content words of the top five turns, a rule-based form of query expansion, lowered multi-hop recall at 40 from 64.9 to 59.5. Greedily selecting 40 turns for relevance minus similarity to turns already chosen changed nothing. Reading deeper for list-shaped questions and shallower for other questions lost to a fixed depth of 40 at the same mean length under every rule tried. The question's shape did not reliably identify the misses.

The ranker's scores did not provide an effective stopping rule either. Reranked scores are flat past rank 40, so every rule that reads until the score drops stopped at its minimum depth. Reading to the largest score gap in the top 150 matched a fixed cut with the same mean length. Capping turns from one session within the top 40 changed nothing at caps of 8 or 12 and lost recall at 4 or 6. The remaining development misses at 40 include 21 multi-hop questions, 10 of them list questions with two to six gold turns spread over sessions.

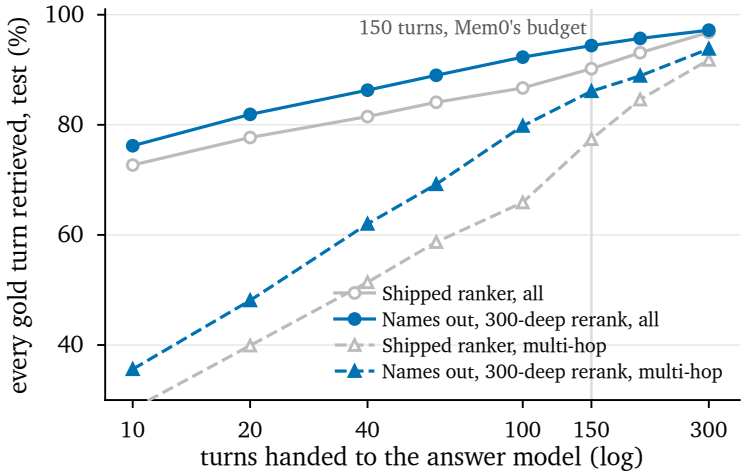


Figure 5: LoCoMo all-gold recall on the 1,152 test questions with evidence labels, against turns handed to the reader, for the shipped graph-free ranker and the same ranker with speaker names kept out of the first stage and the cross-encoders re-scoring 300 candidates. Solid lines are all questions, dashed are the 208 multi-hop questions. Both settings were fixed on the development split and the test split was ranked once.

5.4 LongMemEval

On LongMemEval-S the store sits within a point of Mem0 under a different judge and at more tokens, and therefore we report a band and claim no lead, although a rule that fits Mem0's token budget keeps almost every gold turn. With GPT-5 reading the top 100 turns under Mem0's current answer prompt, the store scores 94.0 on the 319 of 400 test questions the run finished before the budget stopped it, and 93.0 on the 100 development questions, against 93.4 for Mem0's current release on all 500 (467 passed), read from its published file [21]; Mem0's site lists 94.4 for the same release, and we use the file. The judges differ, since ours is the benchmark's official judge on gpt-4o and Mem0's is GPT-5 with its own judge prompt, and our reader sees 19,125 tokens per question against Mem0's 6,787. At 40 turns, the setting of our earlier runs, the reader sees 10,436 tokens and scores 89.0 on development with Mem0's prompt and 87.0 with the official one, and every gold turn is among the 40 for 93.8 of development questions and among the 100 for all of them. The answer split (100 development, 400 test) and the retrieval split (150 development, 350 test) are both prefixes of the same SHA-256 order, so 43 of the 319 finished test answers fall in the 150 questions on which the fusion weight was set; that row therefore carries the tuned mark in Table 1. Mem0's published file carries its judge's verdicts but not its answers, so the two rows cannot be placed under one judge. Re-judging our own answers with Claude Sonnet 5.5 under the official prompts gives 83.0 on development and 87.1 on the 319 test questions, and that judge agrees with the official one on 90.0 and 93.1 percent of them. The band is set partly by the judge.

The token gap closes with a rule about who said the gold turn. On LongMemEval-S the gold turns are almost always the user's, 168 against 13 assistant replies on development, and a user turn averages 53 tokens while an assistant reply averages 358, with about 34 of the top 100 turns being assistant replies, so that most of the 19,125 tokens are replies that hold no answer. A rule that hands the reader every user turn in the top k and an assistant reply only when it ranks in the top 10 cuts the prompt to 6,777 tokens at $k = 60$, within ten tokens of Mem0's 6,787, and keeps every gold turn for 97.9 of development questions, while at $k = 100$ it is 8,712 tokens and 100.0; shortening long assistant replies to their best-matching sentences kept less gold, 96.9, because the answer sentence is often the one cut. On the test split, scored once with the rule fixed, $k = 60$ gives 6,771 tokens and 97.9 and $k = 100$ gives 8,506 and 98.7. A local cross-encoder, bge-reranker-base over the top 150 turns of the first stage, lifts the rule's development recall at 60 turns to 99.0 at 7,026 tokens, 3.5 percent over Mem0's budget, and at 40 turns to 96.9 at 6,029, and on test, scored once, 60 turns give 6,881 tokens and 97.7. The answer run with the reranker and the rule at 60 turns was registered and not run (Section 5.7), so these are offline measurements against the labels with no reader.

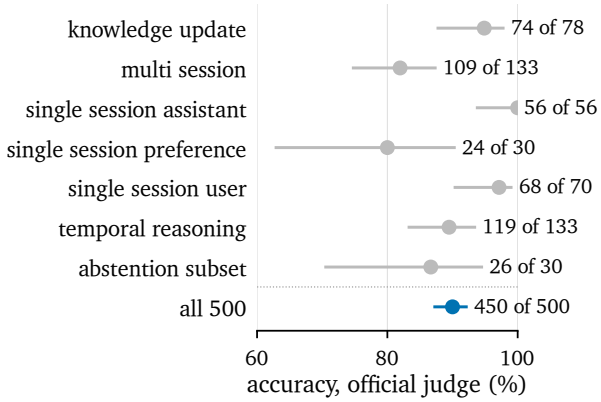


Figure 6: LongMemEval-S answer accuracy by question type, 500 questions, the rank-fusion retriever (5a) at 40 turns with Claude Sonnet 5.5 and the official judge, with 95% Wilson intervals. The abstention row is a subset of the others. No choice was made on this benchmark.

The secondary row with Claude Sonnet 5.5 reading 40 turns under the official prompt scores 90.0 [87.1, 92.3] on all 500 questions (Figure 6), against 85.2 for Supermemory with gemini-3-pro and 71.2 for Zep with gpt-4o, and it answers 74 of 78 knowledge-update, 109 of 133 multi-session, 56 of 56 single-session assistant, 24 of 30 preference, 68 of 70 single-session user and 119 of 133 temporal questions, and 26 of 30 abstention questions. Other published rows are 60.6 for full context with gpt-4o [40], 82.4 for full context with gpt-5-mini [29], 91.4 for Hindsight with gemini-3-pro answering and GPT-OSS-120B writing the memory [14], and 94.9 for Mastra with gpt-5-mini [4, 20], and none of them shares our reader.

5.5 BEAM

On BEAM's 1M-token tier the store and Mem0 are within two points under different judges, and the abilities on which they differ most say where verbatim turns help and where extracted facts do. Over 700 questions on 35 conversations and ten abilities, the store scores a pass rate of 71.8 and a mean rubric score of 0.645 on the 652 questions finished, against 70.1 and 0.641 for Mem0's current release on all 700 [21, 36] (Table 7, Figure 1e). Both use GPT-5 as the reader; ours reads the top 20 turns and, when it answers NOT FOUND, the top 40, which happened for one question in ten, so it sees 22 turns and 13,243 tokens on average. The judges differ, since ours is BEAM's own rubric prompt on gpt-4.1-mini with a question passed when the mean rubric score is at least 0.5, Mem0's rule, and Mem0's is GPT-5 with its own rubric prompt. The benchmark paper's own best system, LIGHT, reaches a mean of 0.336 at this tier with Llama 4 Maverick or GPT-4.1-nano as reader and reports no pass rate, so its rows and these are not comparable. The two systems differ most on abstention, where ours passes 89.4 against 55.7, and on contradiction resolution, 66.7 against 48.6, because a reader looking at verbatim turns can see that nothing supports a claim while an extracted fact list invites an answer; Mem0's row is higher on instruction following, 88.6 against 75.4, on multi-session reasoning, 74.3 against 63.1, and on temporal reasoning, 67.1 against 56.9, and with 65 to 70 questions per ability and different judges these are directions rather than ranked results. The 48 remaining questions were not run. After the budget returned we judged both systems' answers with one other model, Claude Sonnet 5.5 under BEAM's rubric prompt, registered beforehand with a prediction (Appendix A). Under that judge the store passes 61.3 [57.6, 65.0] of its 652 questions with a mean rubric score of 0.558, and Mem0 passes 63.9 [60.2, 67.3] of its 700 with a mean of 0.603. The two sit 2.6 points apart with overlapping intervals, and the order reverses. Both registered ranges, 66 to 76 for ours and 64 to 74 for Mem0, missed on the low side; the registered gap of under 4 points held. The ability pattern survives the change of judge (Table 7): the store keeps contradiction resolution, 63.6 against 7.1, and abstention narrows to 59.1 against 55.7, while Mem0 keeps instruction following, 85.7 against 72.3, multi-session reasoning, 75.7 against 52.3, and temporal reasoning, 68.6 against 56.9.

Table 7: BEAM 1M by ability: pass rate and mean rubric score. Ours on 65 or 66 questions per ability with the GPT-5 reader; Mem0 on 70 per ability with the GPT-5 reader, from its published file. The first two pairs of columns use each system's own judge, BEAM's prompt on gpt-4.1-mini for ours and GPT-5 with Mem0's prompt for Mem0, and are read as a band. The last two pairs re-judge both systems' answers with one model, Claude Sonnet 5.5 under BEAM's rubric prompt, pass at a mean of 0.5.

	Own judges				One judge, Claude Sonnet 5.5			
	Ours		Mem0, current		Ours		Mem0, current	
Ability	pass	mean	pass	mean	pass	mean	pass	mean
Abstention	**89.4**	0.856	55.7	0.525	59.1	0.561	55.7	0.543
Contradiction resolution	**66.7**	0.553	48.6	0.357	**63.6**	0.578	7.1	0.239
Event ordering	64.6	0.569	60.0	0.536	41.5	0.390	51.4	0.483
Information extraction	70.8	0.634	**75.7**	0.700	70.8	0.665	74.3	0.679
Instruction following	75.4	0.726	**88.6**	0.852	72.3	0.687	**85.7**	0.797
Knowledge update	67.7	0.662	65.7	0.650	63.1	0.612	65.7	0.646
Multi-session reasoning	63.1	0.547	**74.3**	0.652	52.3	0.505	**75.7**	0.639
Preference following	93.8	0.862	97.1	0.883	83.1	0.734	**97.1**	0.842
Summarization	69.2	0.639	68.6	0.635	50.8	0.426	57.1	0.534
Temporal reasoning	56.9	0.399	**67.1**	0.618	56.9	0.426	**68.6**	0.629
All (652 / 700)	71.8	0.645	70.1	0.641	61.3	0.558	63.9	0.603

Table 8: Proced-Mem MAP, one value per judge pass. The last row is transcribed from the first Memorable white paper, which used a different lexical retriever. ‡ Chosen on labels from the same 40 queries it is scored on.

Retriever	Pass 1	Pass 2	Pass 3
MiniLM baseline		0.796	
BM25 + bge-small	0.900	0.895	0.821
BM25 + bge-m3	0.932	0.908	–
bge-m3 + IDF overlap‡	0.954	0.949	0.968
First white paper†		0.959 (two passes)	

5.6 Procedures

On Proced-Mem, the retriever of Section 3.6 scores 0.957 mean average precision against 0.796 for the MiniLM baseline under the same gpt-5 judge (Table 8). The retriever was selected using relevance labels from the same 40 queries on which it is evaluated; the benchmark has no held-out split. This result therefore describes the best rule-built setting found on those queries, while transfer to unseen queries remains untested. The baseline is our re-run of the repository's runner; the benchmark paper reports 0.795.

Repeated judging can move the score of a fixed ranking by up to 0.08, so we did not select a retriever from a single judge pass. Instead, we ran an offline sweep of 427 rule-built variants against the benchmark's fixed, judge-confirmed relevance labels, with about 22 relevant trajectories per query. We selected by MAP@10, which tracks the ordering of the judge's MAP scores but is 0.12 to 0.15 lower.

In this offline sweep, task text alone scored 0.798 mean MAP@10, against 0.757 for task plus action text. The bge-m3 embedder scored 0.798, against 0.768 for bge-large and 0.749 for bge-small. IDF-weighted overlap scored 0.782, against 0.759 for stemmed BM25. Changing the fusion constant made no difference, and adding a cross-encoder reranker or pseudo-relevance feedback gave no gain. These scores use fixed labels, so the judge's pass-to-pass spread does not apply; we report no intervals for them. The selected setting averages 0.957 MAP over three judge passes, with precision at one of 0.925, 0.925 and 0.975. Testing transfer requires queries that were not used in the sweep.

5.7 Pending

The model budget ran out before the runs below started, so their registered predictions stand on record without numbers, nothing in this paper is estimated from them, and the addendum numbers refer to entries

in the protocol file. The two registered quantities that could be measured without a model, all-gold recall at 150 and at 40 turns on the test split, are reported in Section 5.3, where the first held and the second landed 0.3 above its range.

- LoCoMo, GPT-5 reader and judge, 40 turns, the 933 test questions left (Addendum 32b). Prediction: test accuracy between 88 and 90.5 against Mem0's 91.1 on the same conversations.
- LoCoMo, token-matched: the search of Section 5.3, 150 turns, one date line per run of same-session turns, 6,522 tokens against Mem0's 6,956 (Addendum 36). Predictions: test accuracy between 90.0 and 92.5 against Mem0's 91.1; multi-hop between 85 and 91.
- LoCoMo, Mem0's 1,540 published answers re-judged with the 2025 judge that scored our Sonnet rows 95.7 (Addendum 35). Prediction: 93.5 to 96.0.
- LongMemEval-S, the 81 test questions left at 100 turns (Addendum 33c; prediction 93 to 95 on 400), and the user-turn rule with the local reranker at 60 turns on development then test (Addendum 36b). Predictions: development between 92 and 95 against the 100-turn run's 93.0; test between 93 and 95.5 against Mem0's 93.4, at about 7,030 tokens.
- BEAM 1M, the 48 questions left, then both systems' answers judged by BEAM's judge and by Mem0's judge (Addendum 36). Predictions: Mem0's answers pass between 62 and 72 under BEAM's judge; ours between 66 and 76 under Mem0's; the ordering holds under both. After the budget returned, both systems' answers were judged by one other model instead, Claude Sonnet 5.5 under BEAM's rubric (Section 5.5, Appendix A).
- LoCoMo and LongMemEval-S with Claude Sonnet 5.5 as reader under Mem0's prompts and GPT-5 as judge (Addendum 35). Prediction: the 150-turn LoCoMo arm between 92 and 94.5 on test.

6 Cost

We measure three costs: the work required to build the store, the input tokens handed to the answer model per question, and the search time from question to ranked list. Building our store requires local encoding and indexing but no generative write calls. On the completed LoCoMo comparison, which mixes development and test questions, the 40-turn setting uses about half as many reader input tokens as Mem0's current release. This saving does not hold for LongMemEval-S as run, and our LoCoMo search is slower than the systems reported in Mem0's 2025 paper. The measurements below state the setup for each comparison. Figure 7 places every LoCoMo row on these three axes at once.

For our runs, read cost counts the reader's whole input, including the instruction and question, with the o200k tokenizer. Mem0's counts come from its own site, so comparisons with those counts are approximate. Under Mem0's prompts and with GPT-5 reading, our 40-turn LoCoMo setting uses 3,294 input tokens per question against 6,956 for Mem0's current release (Table 9, Figure 8). The Sonnet rows with the adaptive cut use 1,263 tokens per question under our own prompt. Mem0's 2025 paper reports tokens injected per query: 1,764 for Mem0, 3,616 for Mem0g, 3,911 for Zep and 26,031 for full context.

Increasing the number of LoCoMo turns raises our input to 4,030 tokens at 60 turns, 5,683 at 100, 7,767 at 150 and 9,895 at 200, counted the same way. Grouping consecutive turns from the same session under one date line, rather than repeating the date on every line, reduces the 150-turn input to 6,522 tokens. This is below Mem0's reported count and is the setting of the token-matched run registered in Section 5.7.

On LongMemEval-S, our 100-turn GPT-5 reader sees 19,125 tokens per question against Mem0's 6,787. The user-turn rule of Section 5.4 brings the 60-turn input to 6,777 tokens on development. On BEAM 1M, our reader sees 13,243 tokens per question; Mem0 does not publish a per-question count.

The build path uses local encoders but makes no generative model call. Embedding the 5,882 LoCoMo turns takes 36 seconds on a laptop CPU. Building a LongMemEval-S history takes 0.19 seconds per unique session; for a LoCoMo conversation, it takes 0.20 seconds per session, or 9 ms per turn. Mem0 makes at least one extraction call and one update call per exchange [3]. EdgeMem meters 0.46M to 11.77M build tokens for six model-built memories on LoCoMo [6].

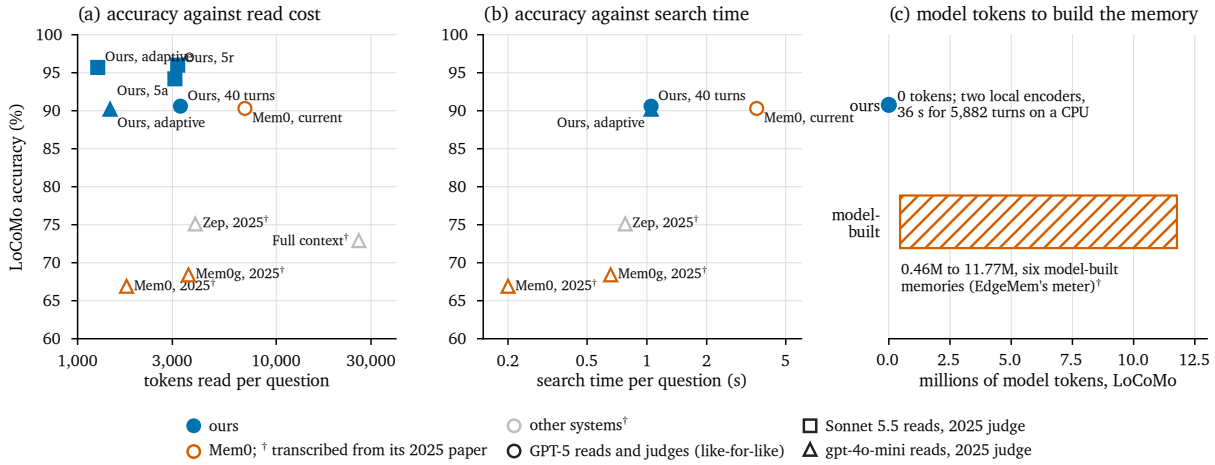


Figure 7: Where the systems land on LoCoMo. (a) Accuracy against tokens the reader is handed per question and (b) against search time per question, both on a log axis; (c) model tokens spent to build the memory. Marker shape is the reader and judge, and only the GPT-5 pair, filled against hollow circles, is like-for-like; † marks rows transcribed from Mem0's 2025 paper, whose tokens are those injected per query and whose times are 95th percentiles on Mem0's hardware, against our medians on one laptop with every cache off. Mem0's current time is the mean its result file logs on its hosted service and includes network. The build range in (c) is EdgeMem's meter of six model-built memories on LoCoMo.

Table 9: LoCoMo per-question read cost and search time. Our build-token counts exclude local encoder and indexing compute and count generative write tokens only. Our search times cover the whole search per question, first stage through both cross-encoders, on one laptop with the score cache bypassed and the models loaded. Mem0's current search time is the mean of the per-question latency in its published result file, on its hosted service. † Transcribed from Mem0's 2025 paper: tokens injected per query, search time at the 95th percentile, and stored size per conversation.

System	Build	Read	Search
Ours, 40 turns, GPT-5	no model calls	3,294	1.05 s, 100 candidates
Ours, 150 turns, grouped	no model calls	6,522	3.16 s, 300 candidates; 1.25 s cascade
Ours, adaptive, Sonnet	no model calls	1,263	1.05 s, 100 candidates
Mem0, current	model	6,956	3.57 s mean
Mem0†	model; about 7k stored	1,764	0.200 s
Mem0g†	model; about 14k	3,616	0.657 s
Zep†	model; over 600k	3,911	0.778 s
Full context†	0	26,031	n/a

Search time depends mainly on reranking. The ranker searches one LongMemEval-S history of about 50 sessions in 27 ms at the median and 31 ms at the 95th percentile, with embeddings stored and ingest excluded. Adding the MiniLM rerank of 24 chunk pairs brings the time to 92 ms.

On LoCoMo, the whole search includes the first stage, the speaker prior and both cross-encoders. It takes 1.05 s per question at the median over 100 candidates (95th percentile 1.19 s), and 3.16 s over 300 candidates (3.38 s). These measurements use 40 development questions, with the score cache bypassed and the models loaded. The first stage alone takes 0.01 s, so cross-encoder reranking accounts for nearly all the search time. The cascade of Section 5.3 reduces this cost by having the small cross-encoder score 300 candidates and the large one score only the top 100. It takes 1.25 s at the median (1.34 s at the 95th percentile). For the secondary retriever 5r, reranking all 1,536 questions with bge-reranker-base on CPU took 1,779 seconds, about 1.2 seconds each.

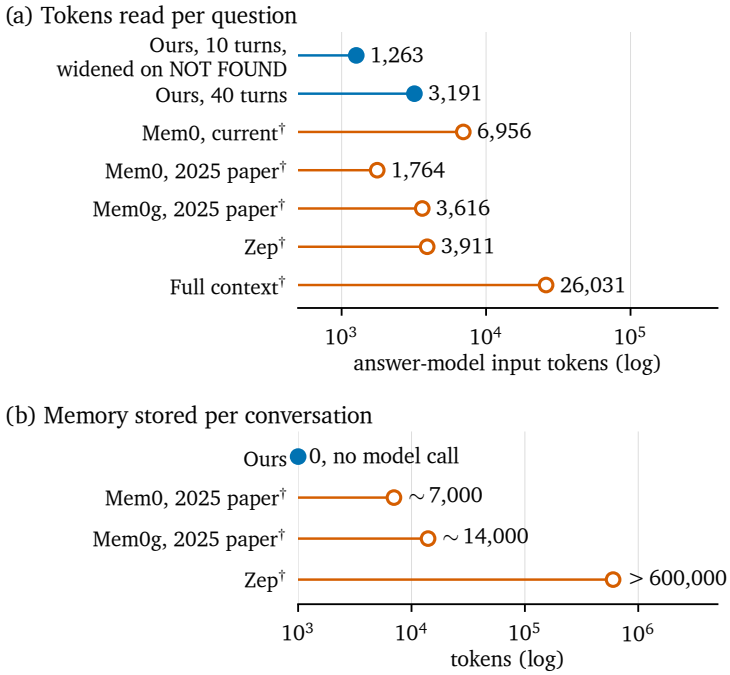


Figure 8: (a) Tokens the answer model reads per LoCoMo question. (b) Size of the memory each system stores per conversation. Our store generates no memory text; it retains the original turns. † Transcribed from Mem0's 2025 paper. Mem0's current release reads 6,956 tokens per question. For Mem0 and Mem0g the figure is the size of the stored facts. For Zep it is the memory store, over 600k tokens, so it is drawn as a lower bound. These are stored sizes and do not count build calls.

These times are not measured under a shared hardware or latency protocol. Mem0's 2025 paper reports 95th-percentile search times of 0.200 s for Mem0, 0.657 s for Mem0g and 0.778 s for Zep on its hardware. Our 100-candidate median of 1.05 s is above all three; the deeper setting and cascade are slower still. Mem0's current hosted service has a logged mean of 3.57 s, above our measured times, but its figure includes network time and is therefore the weakest comparison. Our measurements use one laptop. Peak memory is 8.3 GB on LongMemEval-S and 3.0 GB on LoCoMo.

7 Ablations

We measure component contributions by removing stages from the frozen LoCoMo configuration with the entity graph, using the seven test conversations. The main metric is recall at k: the mean share of gold turns, which the benchmark marks as holding the answer, in the top k. The flat-fusion baseline adds a BM25 keyword score and an embedding score at fixed weights, with no later ranking stages. The two cross-encoders are local models that each read a question and a turn together and produce a relevance score.

Removing both cross-encoders produces the largest measured reduction in recall at 10: 6.1 [+4.5, +7.8] points (Figure 9). Removing contiguity costs 2.5 [+1.4, +3.7] points, and removing the speaker prior costs 0.9 [+0.1, +1.8]. The ranker with the entity graph exceeds the flat fusion by 17.5 [+15.0, +20.2] points at 10, 20.9 at 5 and 9.2 at 50 (Table 10, Figure 10). Cross-encoders help most near the top of the ranking: removing them costs 8.4 points at 5 and 1.2 at 50.

The larger cross-encoder and candidate depth are not separately isolated in the comparison with the first-round chain. That chain reranks the top 50 turns with MiniLM alone. The later ranker gains 4.0 [+2.4, +5.3] points, but it changes both the model and the candidate depth. Within the later ensemble, removing MiniLM changes recall at 10 by −0.1 [−0.9, +0.6], so the smaller model could be dropped without measurable loss.

Reranking is also evaluated in the secondary retrievers. In the answer runs, 5r is 1.8 points above 5a with the same reader and judge: 96.0 [94.7, 97.4] against 94.2 [92.9, 95.7], with overlapping intervals. On

Table 10: Ablation ladder on LoCoMo test, text only, recall at 10 in deployment order. The third row is the graph-free ranker, whose three changes are not separated on test. The fourth row is the frozen configuration with the graph components. The last row has no interval because no paired run of it against the flat fusion was made.

Configuration	@10	Δ@10 vs. flat fusion
Flat fusion	64.0	
+ forward contiguity	68.5	+4.5 [+1.6, +7.2]
+ BM25 weight 0.25, speaker prior, rerank	80.3	+16.3 [+13.8, +19.2]
+ PageRank, entity field, backward contiguity	81.5	+17.5 [+15.0, +20.2]
Ranker without cross-encoders	75.4	+11.4

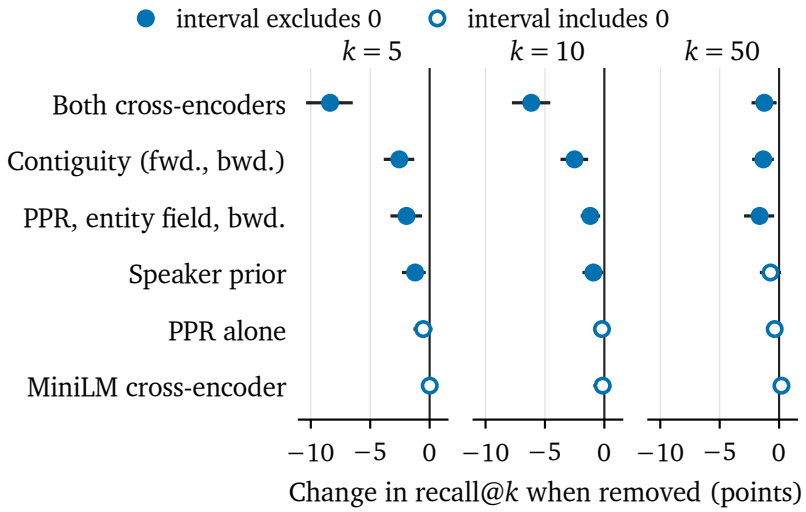


Figure 9: Change in LoCoMo turn recall at k when a part is removed from the ranker with the entity graph (ablated minus full), text only, with 95% cluster bootstrap intervals of the paired differences over the seven test conversations. Rows overlap: the contiguity row and the PageRank row both remove backward contiguity. In the MiniLM row, the weight of bge-reranker-v2-m3 rises to 0.5.

LongMemEval-S, a MiniLM rerank of the top 12 sessions raises test $\text{Recall}_{\text{all}}$@3 to 87.3. Its pooled gain is +1.00 [−0.03, +2.13]; another seed and 4,000 draws give [+0.03, +2.14]. We therefore report this result as secondary and claim no gain.

The deployed store omits the entity graph because its measured contribution is small. On LoCoMo, the entity graph walk, the entity field and backward contiguity together add 0.9 to 1.9 points of turn recall. Removing personalized PageRank alone changes recall at 10 by −0.2 [−0.9, +0.5], which does not rule out a gain under one point. Removing PageRank, the entity field and backward contiguity together lowers recall by 1.2 [+0.4, +2.0] points. Backward contiguity is not an entity link, and the three components were not removed one at a time, so this combined difference cannot be attributed to the graph alone.

On LongMemEval-S, the three components add nothing we could measure. In development, PageRank at weight 0.1 changes the pooled objective by −0.05 [−1.06, +0.96], and adding entity, fact or date terms to the BM25 text changes it by 0 to −0.81. One possible explanation, which we did not test, is that the dense channel already matches the paraphrases an entity link would bridge.

The context-budget experiments measure a tradeoff between accuracy and reader input. Doubling BM25's cut from 20 to 40 turns raises answer accuracy by 3.5 points with GPT-6 Sol reading, from 82.3 to 85.8. Using an adaptive cut that starts at 10 turns, rather than always supplying 40, gives up 0.3 points of accuracy while using 40 percent of the tokens.

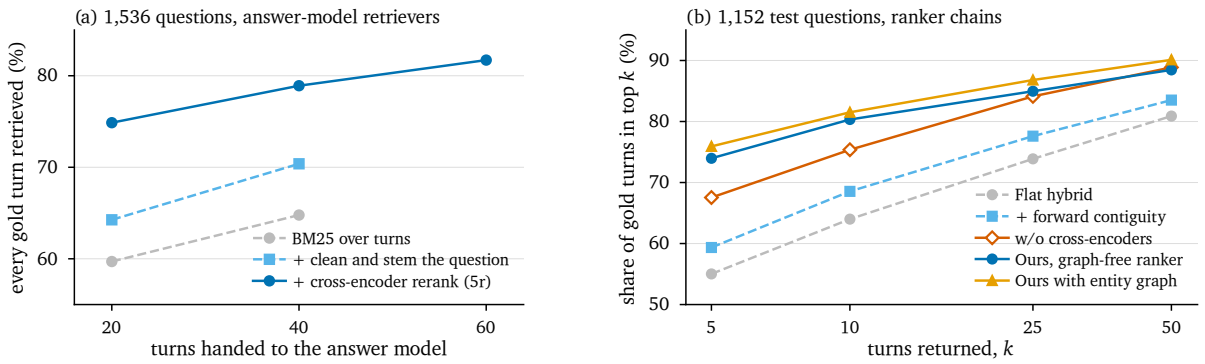


Figure 10: LoCoMo retrieval. (a) The share of 1,536 questions whose gold turns are all among the turns handed to the answer model, one line per step added to the 5a and 5r retrievers. (b) The mean share of gold turns in the top k on the seven test conversations, one line per ranker chain. Panels differ in metric and retriever.

The retrieval experiments compare preprocessing, ranking and retrieval units. At a cut of 40 turns, cleaning and stemming the question raises every-gold-turn recall from 0.648 to 0.704; the dense embedder reaches 0.753, and the reranker reaches 0.789. Adding the neighbour on each side of every hit raises recall to 0.830 but sends about 96 turns, so we did not use that setting. Session-first retrieval lowers recall to 0.583. The bge-m3 embedder recalls less than bge-small on this data, 0.733 against 0.753, and takes longer to embed.

Two additional search changes were evaluated only on LoCoMo development. Pseudo-relevance feedback re-runs the search with words from the top hits. It gains 2.7 points on multi-hop recall, or two of 74 questions, below the registered bar of three. Date windows yield no gain. Neither change was evaluated on test.

8 Limitations

The results support claims about recall over fixed histories under the reported evaluation conditions. The following limits prevent a broader conclusion about the value of generative memory writing.

Incomplete answer evaluations. The matched LoCoMo test comparison covers only 222 of 1,155 questions. On those questions, our store scores 82.0 against Mem0's 85.6, a difference of −3.6 [−7.1, −0.5]. The subset oversamples hard categories. The pooled 607-question result also includes 385 development questions used to tune the ranker, so its aggregate tie does not establish a tie on the full test split. Our LongMemEval-S answer run covers 319 of 400 test questions, and the BEAM run covers 652 of 700. The remaining questions were not evaluated.

Selection on scored questions. Some retrieval results include questions used to choose the configuration. The LongMemEval-S comparison on all 470 questions includes scored questions from the 150-question development split used for tuning. Earlier LoCoMo retrievers 5a and 5r were selected on all questions before the split existed. On Proced-Mem, configuration selection and evaluation use labels from the same 40 queries, and the judge can move a fixed ranking by up to 0.08. The Proced-Mem result therefore does not measure transfer to unseen queries.

Recall gains without answer measurements. The two later search changes were selected on 383 development questions, including 74 multi-hop questions. Removing speaker names from the first stage recovers six development multi-hop questions. The recall improvement also holds on a frozen test ranking, but no reader was run on it. Improved retrieval therefore does not yet demonstrate improved answers. The token-matched LoCoMo run and the LongMemEval-S run with the user-turn rule were registered but never started (Section 5.7); the same-judge comparison was run later, with one Claude judge scoring both systems' existing answers on LoCoMo and BEAM (Sections 5.2 and 5.5).

Reader and judge dependence. Only the main LoCoMo comparison shares Mem0's reader, judge and both prompts. On LongMemEval-S and BEAM the reader is shared, but our benchmark-specific judges differ from Mem0's GPT-5 judge and prompts. These scores cannot rank the two systems. Re-judging both systems' BEAM answers with one model lowers both pass rates by 6 to 10 points and reverses their order while keeping them within 2.6 points. That band is set by the judge as much as by the memory. Our own experiments move one retriever by three points by changing the reader alone, so comparisons across readers also confound the memory with the answer model.

Limits of published comparisons. We did not run Mem0's pipeline ourselves; its scores come from published per-question files, which differ from its site figures by about a point on both chat benchmarks. Our LongMemEval-S retrieval comparison also depends on assumptions about EdgeMem that we have not confirmed with its authors: we interpret its R and N metrics as $\text{Recall}_{\text{all}}$@k and NDCG_{any}@k, and assume its index includes both user and assistant turns. Its per-question outputs are unavailable. The margins therefore treat published results as unrelated samples and do not account for our tuning on part of the scored set. We ran no model-built memory with the same encoders as ours. The published retrieval comparisons therefore mix storage design with encoder choice. On LoCoMo, no published memory reports turn recall in the same form; the 16.3-point gain [+13.8, +19.2] is over our own flat retrieval baseline.

Benchmark defects and uncertainty. Audits report that 6.4 to 7.0 percent of LoCoMo answer keys are wrong, and at least one evidence label in our miss analysis points to an unrelated turn. The LoCoMo test split contains only seven conversations, limiting the precision of conversation-cluster intervals. These issues affect both measured accuracy and interpretations of individual retrieval failures.

Resource tradeoffs. Our 100-turn GPT-5 LongMemEval-S configuration reads 19,125 tokens per question against Mem0's 6,787. The proposed user-turn rule fits Mem0's budget and retains every gold turn for 97.9 percent of development questions, 2 points below the 100-turn setting, but its answer accuracy is unmeasured. On LoCoMo, search takes 1.05 s for 100 candidates, 3.16 s for 300 and 1.25 s with the cascade. These laptop timings are numerically above the 0.200 s 95th-percentile latency in Mem0's 2025 paper, but the hardware and timing summaries differ (Section 6).

Scope. A verbatim store preserves stated evidence but does not add facts absent from the history. Six of the 27 questions Mem0 answers and we miss require world knowledge not stated in a turn or have an evidence label that points elsewhere. The evaluations cover fact recall over fixed histories; they do not measure memory updates, forgetting or agent task success.

9 Conclusion

A memory built from original conversation turns can retrieve useful evidence without a generative writer. On the completed LoCoMo comparison, it reaches similar aggregate answer accuracy to Mem0 while supplying roughly half as many input tokens. That result includes development questions used for tuning, and our system trails Mem0 on the partial test subset. Judging both systems' answers with one other model leaves the tie and the deficit unchanged. It therefore does not establish that a writer is unnecessary for held-out answer accuracy.

The clearest held-out retrieval result is the improvement from 64.0 to 80.3 percent mean gold-turn recall in the top 10 on LoCoMo. The remaining misses show the complementary challenge: preserving a detail does not guarantee that search will find every turn needed to answer. Later search changes improve recall, but their effect on answers remains unmeasured. Completing the held-out, token-matched comparisons is the next test of whether preserving history and retrieving evidence on demand can match model-written memory at the same reader budget.

Reproducibility

The store, the lookup, the figure scripts and the protocol file with every registration and result are in the paper's repository. The split files, the frozen configurations and their hashes, the run logs and the per-question result rows ship with the code. The figure script reads every plotted value from those files, including Mem0's published per-question files, and writes one CSV per figure.

Ethics

Every benchmark used is public, and no new data were collected from people. The reader and judge calls were made to commercial model APIs under the providers' terms, and the spend per run is in Appendix C.

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A Predictions

Every run was registered before it started, with numbered predictions, and Table 11 lists each one and whether it held. Ten missed: three scores landed above the registered range, five below it, one answer-model cost ran over its cap and one judge spread exceeded its bound. Eight registered on 2026-10-30 were never run, because the model budget ran out first; the one-judge re-scoring registered that evening ran once Anthropic credits arrived.

Table 11: Registered predictions. Held means the measured value fell inside the registered range or met the registered comparison.

Run	Prediction	Measured	Outcome
LoCoMo, BM25, 20 turns	70 to 82	82.3	missed by 0.3, high
BM25, 40 turns	at least 2 points over 20 turns	+3.5	held
tuned retriever recall	all-gold recall at 20 turns at least 0.70	0.685	missed
tuned retriever accuracy	at least 87	91.1	held
full context	at least any retrieval pass; at least 10 times the cost	95.7; 14 times	held
5r against 5a, Sonnet	at least 1.5 points	+1.8	held
adaptive cut, ranker with graph	test accuracy within 1.0 of 5r (94.9)	94.8	held
	mean turns read at most 16	12.9	held
	second pass for at most 25 of 100 questions	9.7 of 100	held
	cost at or below 5r ($10.80)	$4.93	held
adaptive cut, graph-free ranker	test accuracy within 1.0 of 94.8	94.5	held
	mean turns within 1.0 of 12.9	13.2	held
	all-gold recall at 10 on test within 1.5 of 74.7	72.7	missed by 0.5
LongMemEval-S, 5a (registered for GPT-6 Sol, run with Sonnet)	76 to 88	90.0	missed, high
	above full context gpt-4o and Zep	90.0 against 60.6 and 71.2	held
	answer model under $10	$13.23	missed
Proced-Mem, bge-m3	0.92 to 0.97	0.920	held, at floor
	judge passes differ by under 0.02	0.024	missed
Registered 2026-10-30 (Addenda 35, 36 and 39)			
LoCoMo, v2 search, 150 turns, grouped dates, test	recall at 150 between 92 and 94.5, multi-hop 83 to 89	94.4; 86.1	held
LoCoMo, v2 search, 40 turns, test	recall at 40 between 83 and 86	86.3	missed by 0.3, high
LoCoMo, cascade rerank, test	recall at 40 within 0.5 of 86.3; at 150 within 0.5 of 94.4	86.1; 93.8	held; missed by 0.1, low
LoCoMo, GPT-5, 40 turns, 933 test questions left	test accuracy 88 to 90.5	–	not run
LoCoMo, v2 search, 150 turns, grouped dates, GPT-5 reader	accuracy 90.0 to 92.5; multi-hop 85 to 91	–	not run
LoCoMo, Mem0's answers under the 2025 judge	93.5 to 96.0	–	not run
LongMemEval-S, GPT-5, 100 turns, 81 test questions left	test 93 to 95 on 400	–	not run

Run	Prediction	Measured	Outcome
LongMemEval-S, local reranker and user-turn rule, 60 turns	dev 92 to 95; test 93 to 95.5	–	not run
BEAM 1M, Mem0's answers under BEAM's judge	pass rate 62 to 72	–	not run
BEAM 1M, our answers under Mem0's judge	pass rate 66 to 76; ordering holds under both judges	–	not run
LoCoMo and LongMemEval-S, Sonnet 5.5 reader, Mem0's prompts, GPT-5 judge	150-turn LoCoMo arm 92 to 94.5 on test	–	not run
Registered 2026-10-30, one judge for both systems (Addendum 41): Claude Sonnet 5.5 re-scores existing answers			
LoCoMo, our 607 answers	88 to 93; Mem0 within 2.0 on the same questions	90.1; Mem0 90.0	held
LoCoMo, Mem0's 1,540 answers	89 to 93	91.3	held
LongMemEval-S, both systems	ours 91 to 95; Mem0 91 to 95; gap under 2.5	–	not run: Mem0's file holds verdicts, not answers
BEAM 1M, both systems	ours 66 to 76; Mem0 64 to 74; gap under 4	61.3; 63.9; gap 2.6	missed, low; missed, low; gap held
agreement of the one judge with the original judge on our answers	at least 90 percent on LoCoMo and LongMemEval-S	LoCoMo 99.0 and 98.6; LongMemEval-S 90.0 and 93.1	held

B Retrieval

The tables below give every retrieval system we ran on the frozen splits, so that the rows in the body can be checked against the baselines they beat, and the labels "z 0.4" and "RRF" name the fusion rule, z-score fusion at BM25 weight 0.4 or reciprocal rank fusion.

	A	B		
System	@5 ↑	@3 ↑	@10 ↑	Pooled ↑
BM25, official script	75.5	69.8	84.0	76.4
BM25, stemmed	88.1	78.9	92.1	86.4
MiniLM, max chunk	86.4	76.1	94.6	85.7
bge-base, max chunk	93.4	83.1	**97.9**	91.4
Contriever (re-impl.)	92.1	–	–	–
EdgeMem-style (re-impl.)	85.1	81.0	91.8	86.0
Hybrid MiniLM, z 0.4	91.7	81.6	96.7	90.0
Hybrid bge, RRF	93.0	81.9	96.1	90.3
Flat fusion (bge, z 0.4)	<u>95.0</u>	<u>84.3</u>	96.7	92.0
Ours	**95.4**	<u>84.3</u>	97.3	<u>92.3</u>
Ours + MiniLM rerank	<u>95.0</u>	**87.3**	<u>97.6</u>	**93.3**

Table 12: All systems we ran on the LongMemEval-S test split, session level, $\text{Recall}_{\text{all}}$@k. Track A has n=302 and scores user turns against the official gold labels. Track B has n=331. Pooled is the mean of the three columns. Here and in Table 16, the best value in each column is bold and the second best underlined.

	A			B
System	@5 ↑	@10 ↑	@50 ↑	@10 ↑
BM25, official script	50.0	59.9	77.5	54.4
BM25, stemmed	64.9	75.2	91.1	70.7
bge-base, max chunk	71.9	83.8	97.7	74.0
Contriever (re-impl.)	66.9	83.1	96.7	–
EdgeMem-style (re-impl.)	62.6	73.2	94.7	61.0
Flat fusion (bge, z 0.4)	**72.8**	**87.1**	**99.0**	**80.1**

Table 13: LongMemEval-S turn-level $\text{Recall}_{\text{all}}$@k on the test split, Track A n=302 and Track B n=331. No published row exists for turn level on S. The flat fusion is the frozen turn-level system; on development it scored 74.4, 83.8 and 97.4 on Track A, and no turn-level variant scored above it.

Table 14: LongMemEval-S retrieval on the frozen split, test questions read once. The test split has 400 questions, and 375 are scored after dropping abstention questions. Session metrics are derived from the turn ranking. R_{any} counts a question when at least one gold item is in the top k; R_{all} when every gold item is. 5r was picked on the 95 scored development questions by turn-level R_{all}@5.

	session level					turn level				
Retriever	R_{any}@5	R_{all}@5	R_{any}@10	R_{all}@10	NDCG@10	R_{any}@5	R_{all}@5	R_{any}@10	R_{all}@10	NDCG@10
BM25 (bm25s)	84.0	68.3	90.4	76.5	65.6	78.4	55.2	85.6	68.8	64.1
bge-small, dense	92.3	76.3	95.7	89.3	64.9	81.9	53.3	90.1	72.3	61.9
z-score fusion, 0.4 BM25 + 0.6 dense	93.6	80.0	97.9	89.9	73.2	86.9	63.2	92.8	78.7	71.0
5a, RRF	95.5	81.6	97.9	92.3	72.2	87.7	61.1	94.7	80.5	70.5
5r, 5a + reranker	**97.6**	**87.2**	**99.2**	**95.5**	**80.9**	**93.1**	**73.6**	**96.0**	**85.1**	**79.1**

Table 15: LoCoMo retrieval on the frozen split. Development: conversations 26, 30 and 41 (383 questions). Test: the other seven (1,153), read once. The pick rule, fixed first, was R_{all}@5 on development, which 5r won.

	development		test			
Retriever	Hit@1	R_{all}@5	Hit@1	R_{all}@5	R_{all}@10	R_{all}@40
BM25	0.329	0.493	0.337	0.469	0.532	0.645
cleaned, stemmed BM25	0.397	0.517	–	–	–	–
bge-small, dense	0.295	0.488	–	–	–	–
5a, RRF	0.373	0.530	–	–	–	–
z-score fusion, best weight 0.4	0.420	0.574	–	–	–	–
5r, 5a + reranker	**0.470**	**0.616**	**0.474**	**0.595**	**0.668**	**0.781**

	Text only				Text plus image caption			
System	@5 ↑	@10 ↑	@25 ↑	@50 ↑	@5 ↑	@10 ↑	@25 ↑	@50 ↑
BM25, official script	29.7	36.6	46.9	53.6	28.9	37.0	47.3	55.2
BM25, stemmed	45.9	54.8	63.9	70.7	45.6	54.2	64.8	71.3
bge-base, max chunk	46.7	55.8	69.4	79.8	44.3	53.1	67.5	77.9
EdgeMem-style (re-impl.)	50.2	62.9	74.5	79.2	49.3	62.4	74.7	79.7
Flat fusion (bge, z 0.4)	55.0	64.0	73.9	80.9	53.9	63.5	74.2	81.4
+ forward contiguity	59.4	68.5	77.6	83.5	58.1	67.7	77.4	84.0
Ranker with graph, no cross-encoders	67.5	75.4	84.1	88.9	66.6	74.8	84.4	89.2
First-round chain	70.9	77.6	85.0	88.9	70.7	77.4	85.3	89.2
Graph-free ranker	74.0	80.3	85.0	88.5	74.9	79.7	85.6	88.7
Ranker with graph	**75.9**	**81.5**	**86.8**	**90.1**	**75.8**	**81.4**	**87.0**	**90.3**

Table 16: Turn retrieval on LoCoMo test conversations 3 to 9 (1,152 questions): mean share of gold turns in the first k. The ranker with graph is the frozen configuration with the entity graph, the entity field and backward contiguity. The graph-free ranker is the store of Section 3. The first-round chain reranks the top 50 turns with MiniLM alone. The other two rerank the top 100 with both cross-encoders. All rows are our own runs through one scorer. No published row is measured the same way.

Change	Metric	Δ
LongMemEval-S development		
Contiguity, turn level	A turn@5	−16.2 [−23.9, −9.4]
PPR at weight 0.1	pooled	−0.05 [−1.06, +0.96]
Entity, fact, date terms	pooled	0 to −0.81
LoCoMo development, flat fusion		
BM25 weight 0.4 to 0.25	recall@10	+2.1 [−0.3, +4.4]
LoCoMo development, removed from the ranker with graph		
Both cross-encoders	recall@10	−6.8
Contiguity (fwd., bwd.)	recall@10	−1.9
Speaker prior	recall@10	−0.9
PPR, entity field, bwd.	recall@10	−0.7
MiniLM cross-encoder	recall@10	−0.4
PPR alone	recall@10	+0.1

Table 17: Experiments run on development only. LongMemEval-S rows change the system named in each row; pooled is the development objective, the mean of Track A $\text{Recall}_{\text{all}}$@5 and Track B $\text{Recall}_{\text{all}}$@3 and $\text{Recall}_{\text{all}}$@10. The last block removes components from the frozen LoCoMo system on the three development conversations; with three clusters we give no intervals. PPR is personalized PageRank.

On the LoCoMo test conversations, recall of every gold turn at 5 by category moves from 0.038 to 0.135 on multi-hop, 0.528 to 0.654 on temporal, 0.164 to 0.274 on open-domain and 0.622 to 0.760 on single-hop, between BM25 and the ranker with the two cross-encoders, and multi-hop is the lowest category.

C Costs

Every dollar spent in this paper went to a reader or a judge, since building the store makes no generative write call, and Table 18 gives each answer run's cost at list prices.

Table 18: Token-priced cost of each answer run, at list prices. Judge tokens are not included, except in the last row, which is a judge-only run.

Run	Model	Cost
LoCoMo, BM25 20 turns, 1,540 questions	GPT-6 Sol	$4.16
LoCoMo, BM25 40 turns	GPT-6 Sol	$7.63
LoCoMo, 5a	GPT-6 Sol	$7.42
LoCoMo, whole conversation in the prompt	GPT-6 Sol	$105.81
LoCoMo, 5a	Sonnet 5.5	$10.51
LoCoMo, 5r	Sonnet 5.5	$10.80
LongMemEval-S, 5a, 500 questions	Sonnet 5.5	$13.23
Proced-Mem, three judge passes	gpt-5 judge	about $2
LoCoMo, BEAM and LongMemEval-S answers of both systems re-judged by one judge	Claude Sonnet 5.5	about $49, from token counts